GENOMIC METHODS TO CHARACTERIZE BREED COMPOSITION AND ENVIRONMENTAL ADAPTATION IN LIVESTOCK

By

MESFIN GOBENA

A THESIS PRESENTED TO THE GRADUATE SCHOOL
OF THE UNIVERSITY OF FLORIDA IN PARTIAL FULFILLMENT
OF THE REQUIREMENTS FOR THE DEGREE OF MASTER OF SCIENCE

UNIVERSITY OF FLORIDA

© 2017 Mesfin Gobena

To family, friends and the universe

ACKNOWLEDGMENTS

I would like to thank my major advisor Dr. Raluca Mateescu for giving me the opportunity to join her lab and for providing financial, intellectual and moral support throughout my master's study. I would also like to thank two other members of my supervisory committee, Drs. Samantha Brooks and Francisco Peñagaricano for the immense support and kindness they have shown me. I also want to thank Drs. Arthur Goetsch and Terry Gipson for their help with the 'Genomics of Resilience in Sheep to Climatic Stressors' study. The tremendous amount of effort their team has spent in collecting live sheep samples from all over US was a good lesson in hard work and resilience. I would like to acknowledge Drs. Carlos Martinez and Mauricio Elzo for their valuable comments and suggestions regarding certain aspects of the breed composition study.

I want to express my heart-felt gratitude for Aselefech Haile, my mother, who sacrificed a lot for me to be here. I would also like to thank my brother Ashenafi Gobena for his financial and moral support throughout my undergraduate and master's studies. I also want to thank the rest of my family and all my friends for their unconditional support.

TABLE OF CONTENTS

page

ACKNOWLEDGMENTS 4

LIST OF TABLES 7

LIST OF FIGURES 8

LIST OF ABBREVIATIONS 10

ABSTRACT 11

CHAPTER

1 PREDICTING BREED COMPOSITION IN AN ANGUS-BRAHMAN-CROSSBRED POPULATION USING GENOMIC DATA 13

Introduction 13
Literature Review 16
Population History of Angus and Brahman 16
Population Structure in a Crossbred Population 19
Methods for Determining Individual Breed Composition Using Genomic Data 21
Model based or parametric methods 22
Nonparametric or distance based methods 24
Selecting a small subset of breed-informative markers 26
Materials and Methods 28
Animal Sampling and Genotyping 28
Identifying Breed Composition Using Whole Genome Data 30
Selecting unrelated animals 30
Model based analysis 31
Principal component analysis 33
Informative Marker Selection and Cross-validation 34
Results and Discussion 36
Genotype Data Quality Control 36
Identifying a Subset of Unrelated Samples 36
Model Based Analysis 37
Principal Component Analysis 41
Informative Marker Selection and Cross-Validation 47
Conclusion 51

2 GENOMICS OF RESILIENCE TO CLIMATIC STRESSORS IN SHEEP 53

Introduction 53
Literature Review 54
Climate, Climate Variability and Climate Change 54

Projections for Future Climate 56
Impacts of Climate Change on Food Security 57
Impact of Climate Change on Livestock Production 58
Maintaining and Improving Environmental Adaptability of Livestock through Genetics 60
Genomics of Adaptive Genetic Variation 61
Materials and Methods 63
Animals and Genotyping 63
Environmental Data 65
Retrieving environmental data 66
Summarizing environmental data 67
Genome-wide Environmental Association Analysis 67
Latent factor mixed model 67
Finding the optimal number of latent factors 69
Gene Ontology Term Enrichment 71
Visualization of Results 72
Results and Discussion 72
Genotype Quality Control 72
Environmental Data Retrieval and Summary 73
Genome-wide Environmental Association Analysis 76
Finding the optimal number of latent factors 76
Environmental association analysis 80
Gene Ontology Enrichment 82
Conclusion 83

R CODES 86

LIST OF REFERENCES 88

BIOGRAPHICAL SKETCH 106

LIST OF TABLES

Table page

2-1 Number of samples per region and breed 63

LIST OF FIGURES

Figure		page
1-1	Histogram showing the distribution of pairwise kinship and divergence estimates by King-robust method among all samples.	37
1-2	Bar plot showing the proportion of the genome contributed by each breed for 74 unrelated samples.	38
1-3	Bar plot showing the proportion of the genome contributed by each breed for 602 samples in the related set.	38
1-4	The scatter plot shows a strong positive relationship between Angus ancestry estimates from ADMIXTURE and Pedigree.	40
1-5	A PC1 versus PC2 scatter plot showing how the first PC agrees with pedigree information.	42
1-6	When plotting the first PC against the second, Brangus cattle coming from at least one generation of Brangus-Brangus mating had more scatter across PC2.	43
1-7	PCA was performed on the unrelated set of animals (blue) and PC1 & 2 values for the rest of animals (orange) were predicted based on their genetic similarity to animals in the unrelated set.	44
1-8	The plot shows the relationship between PC1 and Angus percent estimated by ADMIXTURE for both related and unrelated set of samples.	45
1-9	Each point represents the mean of 25 accuracy values from a 5-fold cross validation replicated 5 times performed on a single set of SNP.	48
1-10	The plot illustrates the drop in the genome-wide average MAF as the Angus percent decreases from >90% (A) to <10% (J).	50
2-1	The US map showing the sampling locations of sheep in the current study	64
2-2	A map showing sampling locations and locations of stations from which data was retrieved.	73
2-3	A heat map showing the relationship between the 5 environmental variable considered..	74
2-4	A scatterplot sowing the position of the sampling locations respective to the first 2 top PC.	75

2-5 A scatterplot showing the result from the cross-validation procedure of sNMF run on 8 different values of K. 76

2-6 Histogram showing pairwise KING-robust kinship and divergence estimates for all sheep. 78

2-7 A PC1 versus PC2 scatterplot from PCA applied to genotype data showing overall population structure among all sheep in the study (n=181). 79

2-8 Histogram showing distribution of p-values from LFMM. The uniform distribution of neural (high p-value) loci is indicative of a well-calibrated genomic scan for adaptive loci (Francois et al., 2016) 81

2-9 Manhattan plot showing negative log of p-values from LFMM for loci across the genome. For each chromosome, one SNP with the highest negative log of p-value was labeled with its variant ID. 82

LIST OF ABBREVIATIONS

Fst	Wright's Fixation Index
LD	Linkage Disequilibrium
PCA	Principal Component Analysis
PC	Principal Component
MDS	Multi-Dimensional Scaling
IBS	Identity by Descent
IBD	Identity by Descent
SNP	Single Nucleotide Polymorphism
DNA	Deoxyribose Nucleic Acid
QC	Quality Control
MAF	Minor Allele Frequency
HWE	Hardy-Weinberg Equilibrium
LFMM	Latent Factor Mixed Model
GIF	Genomic Inflation Factor
VEP	Variant Effect Predictor
EA	Environmental Association
IPCC	Intergovernmental Panel for Climate Change
ENSO	El Niño/Southern Oscillation
NAO	North Atlantic Oscillation
RF	Radiative Forcing
GHG	Green House Gases
RCP	Representative Concentration Pathway

Abstract of Thesis Presented to the Graduate School
of the University of Florida in Partial Fulfillment of the
Requirements for the Degree of Master of Science

GENOMIC METHODS TO CHARACTERIZE BREED COMPOSITION AND ENVIRONMENTAL ADAPTATION IN LIVESTOCK

By

Mesfin Gobena

August 2017

Chair: Raluca Mateescu
Major: Animal Sciences

The thesis includes two studies intended to help meet the challenge of finding an optimum balance between productivity and environmental adaptability in livestock. In the first chapter, the goal was to evaluate the feasibility and accuracy of using genomic data to determine breed composition in Angus-Brahman crossbred cattle genotyped with a high-density SNP chip. After applying a series of quality control filters, 54,728 SNP and 676 cattle remained and were used in subsequent analysis. Population structure was characterized by applying a maximum-likelihood model method and principal component analysis to genotype data. Subsets of breed-informative SNP were also selected using pairwise Fst values. There was a strong agreement between breed composition estimated from genotype and pedigree (R=0.96), although there were discrepancies between the two for certain animals. A distinct pattern of variation in cattle with extended Brangus lineage was also observed. Using as few as 15 breed-informative SNP, it was possible predict breed composition with high accuracy (0.95). In the second chapter, the goal was to identify loci affecting environmental adaptation using 184 sheep sampled from four regions of the US with divergent climatic conditions.

Genotyping was performed using the OvineSNP50 BeadChip. Climatic condition of the sampling locations was characterized by summarizing 20 years of daily data for five environmental variables. Loci associated with environmental variables were identified using Latent factor mixed model. After controlling for false discovery rate, 389 SNP were identified as being significantly associated with environmental variation, and these were co-located with 184 genes.

CHAPTER 1
PREDICTING BREED COMPOSITION IN AN ANGUS-BRAHMAN-CROSSBRED POPULATION USING GENOMIC DATA

Introduction

Around 40% of all beef cattle in the US are located in the subtropical Southern and Southeastern parts of the country (Cundiff et al., 2012). Since beef cattle are mostly managed in outdoor conditions, they are prone to face a significant level of exposure to climatic elements (Walthall et al., 2012). Taurine beef cattle breeds of European origin such as Angus, while being highly productive in temperate areas, are not well suited for such climatic zones due to their history of adaptation to temperate regions (Burrow, 2015). In tropical areas, the combined effect of high ambient temperature, increased abundance of parasitic and parasite transmitted diseases, and nutritionally poor pastures leads to poor growth rate and reduced reproductive performance in these cattle (Burrow, 2015).

On the other hand, beef cattle breeds of Zebu lineage such as Brahman, are well adapted to tropical and subtropical environments, owing to the fact that they have evolved in areas with such climatic conditions (Lenstra et al., 2014; Burrow, 2015). Zebu cattle have several characteristics that enable them to withstand the challenges imposed by harsh tropical climates. These include: lowered metabolic rate, which reduces heat production; increased capacity to sweat and larger skin surface area, which facilitates heat dissipation; reduced susceptibility to parasitic diseases and efficient utilization of low quality pastures (Lenstra et al., 2014; Burrow, 2015). However, Brahman cattle, similar to most other Zebu, lag behind Taurine breeds in production traits such as growth rate, reproductive performance and carcass quality (Gaughan et al., 2010).

One way to enhance beef production in tropical and subtropical areas is to use cattle that are crossbreds between European Taurine and Zebu breeds (Fallis, 2012). These crossbreds combine the production performance of Taurine cattle with the tropical adaptation of Zebu cattle, and usually outperform purebred cattle from the parental breeds in subtropical conditions due to heterosis (Burrow, 2015). Angus-Brahman crosses are typically better suited for beef production than other Zebu-Taurine combinations in subtropical parts of the US (Chase et al., 2004). Whether a particular Taurine-Zebu combination is optimal depends on the specific production environment under consideration. It has been suggested (Cundiff et al., 2012) that, in subtropical regions of the US such as the Gulf Cost, cattle with a 1:1 Taurine to Zebu ratio would be preferred whereas a 3:4 Taurine to Zebu ratio would be better suited to the more northern but still subtropical parts of the US such as Southeastern Oklahoma and most of Texas.

Having accurate knowledge of breed composition is essential in evaluating the adaptability of crossbreds to a given production environment (L A Kuehn et al., 2011). Pedigree information is conventionally used to determine breed composition in crossbred cattle (Frkonja et al., 2012a; Vanraden and Cooper, 2015). However, the reliability of pedigree based estimation of breed composition can be compromised by missing, inaccurate or incomplete records (Vanraden and Cooper, 2015). In addition, Mendelian sampling during gametogenesis can lead to deviations from breed composition expected from the pedigree (L A Kuehn et al., 2011).

Using genomic data to determine breed composition offers multiple advantages over using pedigree records. Breed composition derived from genomic data was shown

to be more accurate whilst not being prone to missing, incomplete or inaccurate records (L A Kuehn et al., 2011; Dodds et al., 2014; Funkhouser et al., 2016). Another use of genomic information could be independent authentication of breed in breed-labeled beef products (Wilkinson, 2012).

Disadvantages of using genomic data may include genotyping cost and the need for more advanced technical expertise. Both of these drawbacks can be offset by the fact that genetic and genomic methods are becoming more widely adopted and accessible (Wiggans et al., 2011). The increasing availability of core sequencing facilities at academic and research institutes, combined with the availability of affordable genotyping services from biotech companies, are likely to improve the accessibility and feasibility of using genomic information to determine breed composition (Gould, 2015; Bauck, 2016).

Genotyping cost can also be further reduced by only genotyping breed-informative markers (Wilkinson et al., 2011b). Using a small number of carefully selected breed-informative markers is also advantageous in that it minimizes statistical noise coming from other markers whose frequency has been affected by demographic events that are not relevant to breed membership inference (Wilkinson et al., 2011b).

The goal of the current study was to evaluate the feasibility and accuracy of using genomic data to determine breed composition. The study was performed using 782 Angus-Brahman crossbred cattle with genotype and pedigree information. The objectives of the study were to:

1) Use genomic data to detect population structure due to differences in breed composition by means of parametric and non-parametric methods while accounting for the confounding effect of close familial relationships;

2) Compare breed composition inferred from genomic data with breed composition derived from pedigree;

3) Select a small number of breed-informative genetic markers that can be used to identify breed composition without significant loss of accuracy.

Literature Review

Population History of Angus and Brahman

The level of genetic diversity in domestic cattle seen today is a consequence of several evolutionary forces. Taurine cattle were domesticated from their wild ancestor *Bos primigenius taurus* around 8,500 BC in the Southwest Asian Fertile Crescent (Lenstra et al., 2014) whereas Zebu cattle were domesticated from their wild ancestor *Bos primigenius namadicus* around 6,000 BC in the tropical Indus Valley (Ajmone-Marsan et al., 2010; Lenstra et al., 2014). The dispersal of domestic cattle along with migratory farmers led to the development of different ecotypes that have adopted to a variety of local environments (Decker et al., 2013; Lenstra et al., 2014). Introgression from local wild aurochs, which already had wide geographical distribution, also contributed to local adaptation by early migratory domestic cattle (Vilà et al., 2005; Verhoeven et al., 2011; Lenstra et al., 2014). There were also multiple instances where early Taurine and Zebu populations crossed paths and mixed during migration events, forming crossbred populations (Ajmone-Marsan et al., 2010).

Domestication had a tremendous effect on the genetics, morphology, physiology and behavior of cattle. Compared to their wild ancestor, cattle became tamer and smaller,

and have smaller or no horns (Ajmone-Marsan et al., 2010; Lenstra et al., 2014). Zebu cattle gained their characteristic hump only after domestication (Lenstra and Felius, 2015). Earlier days of domestication saw the development of several ‘agro-types’ which varied in coat color, productivity and environmental adaptation (Lenstra et al., 2014). Since the 18th century, more systematic breeding, mostly involving Taurine cattle, resulted in the formation of hundreds of specialized breeds adapted for a variety of purposes and to a variety of environmental settings (Felius et al., 2011). Taurine breeds are well adapted to temperate conditions, although there are prominent exceptions such as N’Dama, whereas Zebu type breeds are suited for tropical and subtropical climates (Chan et al., 2010).

Aberdeen-Angus or simply Angus, is a breed of Taurine cattle that is known for its high-quality beef. Angus cattle are polled and have either black or red coat color, although red colored Angus are considered a separate breed in the US (Briggs and Briggs, 1980a). The development of Angus as a breed started in late 18th century in Northeastern Scotland which is located further north than the contiguous United States and has a temperate climate (MacDonald and Sinclair, 1910). The breed was formed as a cross between two polled strains: Angus dodies and Buchan humlies (Briggs and Briggs, 1980a). Selection criteria included polledness, coat color, size, symmetry and tendency to accumulate flesh (Briggs and Briggs, 1980a). The Polled Cattle Society, now called The Aberdeen-Angus Cattle Society, was established in Scotland in 1879, although the Herd Book was started much earlier in 1862 (MacDonald and Sinclair, 1910; Briggs and Briggs, 1980a). The first registered Angus were imported into the United States in 1878, and The American Angus Breeder’s Association was established in 1883 (Grey, 1919).

The breed quickly became popular in the US for its rapid growth, high dressing percentage, quality beef, and ability to thrive under winter conditions (MacDonald and Sinclair, 1910; Sheets, 1915). As of September 2016, 334,607 head of Angus cattle were registered in the US (American Angus Association, 2016). However, similar to most other Taurine cattle selected and developed in temperate regions, Angus cattle struggle to maintain production performance in parts of the US with subtropical and tropical climate (Garrick and Ruvinsky, 2015). In contrast, Zebu cattle have evolved in tropical and subtropical conditions and are well adapted to this environment (Lenstra et al., 2014).

Brahman is a beef cattle breed that is of mainly Zebu origin and is known for its tropical adaptation (Buchanan, D S, Lenstra, 2015). It is characterized by a large hump, a well-developed dewlap and a coat with varying shades of grey and red (Akerman, 1982). The breed was developed in the US by mixing four strains from India, namely Guzerat, Nellore, Gir and Krishna valley, although several breeds of European origin also contributed (Briggs and Briggs, 1980b). Close to 300 cattle of Indian origin, most of which were bulls, were imported either directly from India or from Brazil and Canada between the middle of 19th and 20th centuries (Briggs and Briggs, 1980b). Initially, these cattle were mainly used for draught power, but were latter crossed with local breeds with the intention of utilizing their adaptive qualities (Akerman, 1982). The need to have a constant source of cattle of Indian origin for the purpose of crossbreeding led to the establishment of ‘pureblooded’ herds that were blends of the four Indian strains (Akerman, 1982). The American Brahman Breeders Association was organized in 1924, which cemented the position of Brahman as a separate breed and led to its improvement through the introduction of breeding standards (Briggs and Briggs, 1980b; Akerman, 1982). Despite

their advantage in terms of adaptation to hot climate, Brahmans have less desirable characters such as poor reproductive performance, slow maturity, and poor meat quality (Turner, 1980).

Angus and Brahman are complementary in that crosses between them combine the meat quality and reproductive performance of Angus with the tropical adaptation of Brahman (Turner, 1980; Chase et al., 2004). Such crosses are common in Southern and Southeastern parts of the US (Riley et al., 2007). Crossbreeding takes advantage of both additive effects and dominance effects or heterosis (Long, 1991; Burrow, 2015). It has been suggested that, in general, a 50:50 Taurine: Zebu ratio is an optimum combination for the hot and humid Gulf Coast states of the US whereas a 25:75 ratio is suited for the more northern subtropical areas such as Texas and Southeastern Oklahoma (Cundiff et al., 2012). Brangus is a composite breed that is made up of 5/8 Angus and 3/8 Brahman (Briggs and Briggs, 1980b).

Population Structure in a Crossbred Population

Genetic population structure (simply referred to as population structure henceforth) in a given population can be described as the presence of distinct subgroups with characteristic allele frequencies with regard to certain genetic variants (Wright, 1951; Pritchard and Rosenberg, 1999). Such structure does not exist in a panmictic population, but develops as a result of nonrandom mating which can be caused by geographical barriers or selective breeding in the case of domestic species (Wright, 1951). Depending on the extent of nonrandom mating, rate of migration between subpopulations, number of generations elapsed, level of genetic variation, intensity of selective pressure and size of the total population and/or subpopulations, subsequent differentiation among subgroups can occur (Roughgarden, 1979;

Zhivotovsky, 2015). The main driving force behind differentiation is genetic drift, especially for small population sizes, but selective pressure can also play a part if it is strong enough (Roughgarden, 1979). One consequence of such differentiation is an increase in the number of homozygote genotypes and deviation from Hardy-Weinberg Equilibrium (HWE) in the overall population, even as HWE is maintained in subpopulations as described by Wahlund (Termed Wahlund's effect; Sinnock, 1975). For a given locus, the level of differentiation among subgroups can be measured using the Fixation Index (Fst), a parameter that reflects the proportion of heterozygotes in subpopulations relative to the whole population, when no structure is assumed (Wright, 1951). Wahlund' s effect is reversed when differentiated subpopulations are admixed, leading to an elevated number of heterozygotes in the total population (Zhivotovsky, 2015). Such admixture events also increase the extent of linkage disequilibrium (LD) in admixed groups (Pfaff et al., 2001; Thornton et al., 2012), although the extent of LD diminishes as there is more mixing.

The genetic structure of a population can be used to make inferences about its evolutionary history and contributions from different ancestral populations or breeds (the terms *breed* and *ancestral population* are used interchangably henceforth; Rosenberg 2002; Shringarpure and Xing 2014). Identifying population structure in a given population is typically approached as a clustering problem, where the attempt is made to identify subpopulations (clusters) with distinct allele frequencies (Pritchard et al., 2000). Purebreds have distinct allele frequencies at multiple loci as a result of evolutionary forces that led to differentiation. A crossbred population is expected to have allele frequencies that are a linear combination of allele frequencies at corresponding loci in the parental

populations, weighed by the proportional contribution each parental population (Long, 1991; Patterson et al., 2006). However, deviations from this expectation can occur due to sampling bias when computing allele frequencies or chromosomal sampling (i.e., genetic drift; (Long, 1991; Patterson et al., 2006).

In a crossbred population, differences in allele frequencies due to population structure as a result of heterogeneous breed ancestry can be used to make inferences about breed membership (Long, 1991; Pritchard et al., 2000; L. A. Kuehn et al., 2011). The inference can be made on an individual or population level (L. A. Kuehn et al., 2011; Padhukasahasram, 2014). Inference about individual breed membership can also be made locally, which involves assigning breed of origin to chromosomal segments probabilistically, or globally, by estimating proportional contribution from parental breeds averaged over the entire genome (Porras-Hurtado et al., 2013).

Methods for Determining Individual Breed Composition Using Genomic Data

Different methods have used allele frequencies of genome wide markers to identify population structure due to differences in breed composition or ancestry (Pritchard et al., 2000; L. A. Kuehn et al., 2011). Such analyses are commonly performed along with genome wide association studies with the aim of reducing spurious associations by accounting for population stratification (Pritchard and Rosenberg, 1999; Price et al., 2010). Both parametric and nonparametric or distance based methods have been used for this purpose. This section focuses on both parametric and nonparametric methods for inferring individual genome-wide breed ancestry.

Model based or parametric methods

Parametric methods assume the genotypes of an individual from a particular subgroup are random draws from a model with parameters (i.e. allele frequency estimates for the loci being considered) unique to that subgroup, with the expected number of subgroups specified *a priori* (usally designated with the letter K; Pritchard et al. 2000). Given genotype data, subgroup membership and expected allele frequencies for each subgroup are estimated using either Bayesian or maximum likelihood approaches (Liu et al., 2013; Padhukasahasram, 2014). One Bayesian method (implemented by the software STRUCTURE) estimates group membership coefficients for each sample and representative allele frequencies for each group simultaneously. This is done by constructing their joint posterior probability distribution, given genotype data, and sampling from this distribution to come up with estimates using Markov Chain Monte Carlo (MCMC; Pritchard et al. 2000; Raj, Stephens, and Pritchard 2014). Another Bayesian approach that is used by the software fastSTRUCTURE estimates the model parameters using an optimization method called Variational Bayesian inference instead of MCMC, which makes it computationally more efficient (Raj et al., 2014).

On the other hand, maximum likelihood methods use different optimization algorithms such as expectation maximization (implemented by the software Frappe; Tang et al. 2005) and block relaxation (implemented by the software ADMIXTURE; Alexander, Novembre, and Lange 2009) to find the most likely values for subgroup allele frequencies and subgroup membership coefficients given genotype data. All the models mentioned above allow estimation of fractional group membership which is important for determining the level of admixture and hence breed composition for individual genomes (Pritchard et al., 2000; Tang et al., 2005; Shringarpure et al., 2016).

Model based methods have numerous caveats due to the necessity to make certain assumptions, which are not always met. One such assumption is that sampled individuals are representative of their respective populations or subpopulations. As a result, sampling bias leads to inaccurate inference (Pritchard et al., 2000; Shringarpure and Xing, 2014). A corrective method to account for sampling bias has been suggested by Shringarpure and Xing (2014). Another assumption inherent to these models is that there are no close familial relationships between samples included in analysis (Pritchard et al., 2000; Shringarpure et al., 2016), as this can confound ancestry estimates (Matthew P Conomos et al., 2016). The software ADMIXTURE offers a work-around to accommodate related samples by determining structure for the largest subset of unrelated samples and then projecting the genotypes of the rest of the samples on this structure to obtain estimates (Shringarpure et al., 2016).

In addition, most methods also assume linkage equilibrium between markers, which is usually not the case in high density genotype datasets (Raj et al., 2014). The presence of widespread LD, which is more pronounced in recently admixed populations, negatively affects the performance of most model based methods (Patterson et al., 2006). The effect of high LD can be moderated by applying LD pruning on genotype data before analysis (Shringarpure et al., 2016), and/or using a model that accounts for LD (Pritchard et al., 2000). On the other hand, methods that infer local ancestry, such as implemented by the software fineStructure, take advantage of LD and use the information to identify haplotype blocks for which ancestral origin is determined, leading to more accurate ancestry estimates (Lawson et al., 2012).

Another challenge when it comes to model-based approaches is computational burden, especially when applied to large datasets (Patterson et al., 2006; Raj et al., 2014). However, this is becoming less of a problem with the advent of powerful computers and improved algorithms such as the ones implemented by the programs ADMIXTURE and fastSTRUCTURE (Raj et al., 2014; Shringarpure et al., 2016).

Nonparametric or distance based methods

Nonparametric methods to detect population structure are different from parametric methods in that they do not have explicit model assumptions (Pritchard et al., 2000). Such methods typically involve usage of principal component analysis (PCA) alone or together with different cluster analysis tools (Padhukasahasram, 2014). Principal component analysis is a dimensionality reduction technique applied to high dimensional genotype data to identify major axes of variation that capture most of the underlying structure (Jolliffe, 2002). It involves applying eigendecomposition to a marker covariance matrix to identify eigenvectors and eigenvalues. Eigenvalues represent axes of variation that are orthogonal to each other, relative to which the data can be described. Principal component (PC) is a term closely related to, and often used interchangeably with eigenvector. It refers to the projection of genotype data onto the axis of variation associated with an eigenvector. A PC can also be described as a latent variable that is a linear combination of genotype values in columns of the genotype data. An eigenvalue represents variance of data along an eigenvector, and hence, variance of the associated PC (Jolliffe, 2002; De Iorio et al., 2015). The top few PCs that capture the majority of the variation in the genotype data are typically used in plots to visualize genetic similarity between samples (Patterson et al., 2006). Subgroups with similar ancestry are expected to form clusters in these plots whereas admixed

individuals lie along the line between clusters of the parental groups (Patterson et al., 2006; Lawson et al., 2012).

Multidimensional scaling (MDS) is a group of techniques closely related to PCA that map pairwise genetic distance to coordinates on lower-dimensional space so that similarities between samples can be easily visualized (Borg and Groenen, 2005). An important distinction is that, while PCA retains a given number of important PCs, non-metric MDS fits the distance matrix to a specified number of dimensions where the pairwise distances are in a Euclidian space, which leads to a more accurate visual representation (Borg and Groenen, 2005). Eigenstrat and PLINK are software commonly used to perform PCA and MDS on genomic data analysis, respectively (Liu et al., 2013).

Cluster analysis refers to a group of non-parametric methods that involve constructing pairwise distances between samples based on genetic data, and looking for clusters of individuals with similar genetics (Lawson and Falush, 2012). Pairwise distance, measured by Identity by State (IBS) or similar metrics, is used to construct a distance matrix which serves as an input for the clustering step, with or without applying the dimensionality reduction (Lawson and Falush, 2012). Clustering is performed by means of iterative hierarchical (e.g., AW-clust) or non-hierarchical (e.g., k-means) algorithms, with or without specifying the number of clusters *a priori* (Liu and Zhao, 2006; Gao and Starmer, 2007; Lawson and Falush, 2012; De Iorio et al., 2015). These algorithms are implemented in PLINK1.9 as well as various R packages (R Core Team, 2013).

Similar to model based methods, close familial relationships confound identification of population structure due to distant ancestry in PCA (Patterson et al., 2006; Shringarpure et al., 2016), one of the most commonly used nonparametric methods. One approach used to minimize this confounding, if the sample size is large enough, is to find a subset of unrelated samples and use these as a reference population. After identifying structure in the unrelated set, ancestry estimates for the rest of the samples is obtained by projecting their genotype onto this structure (Matthew P. Conomos et al., 2016; Shringarpure et al., 2016). This methodology has been implemented in the R package Genesis (Matthew P. Conomos et al., 2016; Shringarpure et al., 2016).

Distance based methods are more of an exploratory data analysis tool and, it can be difficult to assess the meaningfulness of the classification and derive statistical inference, although there has been significant progress in that aspect (Pritchard et al., 2000; Alexander et al., 2009; McVean, 2009). Moreover, they also do not perform as well as model-based methods in the presence of large linkage blocks as a result of recent admixture (Patterson et al., 2006). However, distance based approaches offer numerous advantages over model-based methods. These include allowing better visualization of patterns of genetic variation (Raj et al., 2014) and being generally computationally more efficient (Patterson et al., 2006; Gao and Starmer, 2007; McVean, 2009).

Selecting a small subset of breed-informative markers

Genome wide single nucleotide polymorphism (SNP) data has been used to accurately determine breed composition in crossbred animals (L. A. Kuehn et al., 2011; Funkhouser et al., 2016). Identifying a small number of genetic markers, usually termed

informative markers, that can be used to predict or estimate breed composition without loss of accuracy can reduce genotyping cost (Rosenberg et al., 2003). In addition, in population genetic studies, selection of a minimum number of informative markers can reduce noise due to uninformative markers (Wilkinson et al., 2011a). Various methods to select informative markers are proposed in a number of studies. These include: absolute allele frequency difference or delta (δ), informativeness for assignment (I_n), pairwise Wright's Fst, and PCA loadings (Rosenberg et al., 2003; Paschou et al., 2010a; Wilkinson et al., 2011a). The first two methods are closely related to Wright's pairwise Fst (Wilkinson et al., 2011b).

A series of articles by Paschou et al (Paschou et al., 2007; Paschou et al., 2008; Paschou et al., 2010b; Lewis et al., 2011) have shown that a small number of markers with strong association to major axis of variation in the genotype data, as identified by PCA, can be used to trace ancestry in complex admixed populations. The methodology used in these studies consists of selecting markers based on the sum of their loading coefficients for top significant PCs, followed by hierarchically assigning individuals to populations and sub-populations using Nearest Neighbors algorithm with distance defined by IBS. Using this approach, it was possible to accurately assign individuals sampled from 51 populations around the world (Cann et al., 2002) to their respective populations (Paschou et al., 2010b). The same procedure was also successfully used to assign cattle from 19 breeds to their respective breeds, although assigning fractional breed membership or composition was not considered in this study (Lewis et al., 2011). Paschou et al (2007) also reported that this PCA-based method performed better than Fst-based methods in identifying markers with the most information on population

structure or ancestry. However, since PCA identifies informative SNPs relative to other SNPs included in the analysis, different SNP can be identified as informative from different SNP sets (Wilkinson et al., 2011b).

Fixation index measures the level of genetic differentiation between subpopulations (Wright, 1951). It is calculated for individual loci and ranges between zero, which means no differentiation, to one, which means complete fixation of alternative alleles in the respective subpopulations (Wright, 1951). A commonly used way to estimate Fst is a method by Weir and Cockerham (1984) which accounts for sample number difference between the populations being compared.

It has been shown (Rosenberg et al., 2003; Wilkinson et al., 2011b; Frkonja et al., 2012a) that a small number of informative markers with high pairwise Fst can be used to differentiate between breeds. The number of markers needed for accurate breed assignment is a function of the level of differentiation between the breeds – the larger the amount of differentiation, the smaller the number of informative markers needed (Patterson et al., 2006; Wilkinson et al., 2011b). Wilkinson et al (2011b) reported that Fst out-performed δ, I_n and PCA in selecting informative SNP used to assign cattle from 17 breeds to their respective breed of origin. In a similar study by Frkonja et al (2012a), as few as 48 SNP selected based on Fst were sufficient to differentiate between two taurine cattle breeds with an accuracy of 0.9.

Materials and Methods

Animal Sampling and Genotyping

A total of 782 animals sampled from the Multibreed Angus-Brahman herd at the University of Florida were used in this study (Elzo and Wakeman, 1998). The herd was constructed using a diallel crossbreeding scheme where six groups of sires with

different proportions of Angus and Brahman, as determined from pedigree, were reciprocally mated with 6 dam groups which were classified in the same manner as the sires (Komender, 1988; Elzo and Wakeman, 1998). The six sire/dam groups were: group one (> 4/5 Angus); group two (3/4 Angus and 1/4 Brahman); group three (5/8 Angus and 3/8 Brahman); group four (1/2 Angus and 1/2 Brahman); group five (1/4 Angus and 3/4 Brahman) and group six (> 4/5 Brahman). The progeny coming from the diallel matings were again classified into six groups using the same criteria as the sire/dam groups. The animals included in the current study were sampled to be representative of all six sire/dam/progeny groups and consisted of 126, 120, 123, 159, 84 and 170 cattle from groups one to six, respectively.

Genomic DNA was extracted from blood samples in three main steps using the QIAGEN® DNeasy® kit (QIAGEN, 2006). The first step involved lysis of the cells in samples by mixing 100 µL of blood with 20 µL of proteinase K in a 2ml Eppendorf tube and incubating at 56 ºC for 10 minutes. In the second step, the lysate was transferred to a DNeasy® mini spin column and centrifuged, during which the silica-based membrane in these columns captured DNA molecules while other components of the lysate passed through. Remaining impurities were removed in two subsequent washing steps. In the last step, the DNA bound to the silica-based membranes was eluted with a buffer solution (10 mM Tris·Cl & 0.5 mM EDTA). The DNA samples were genotyped using GeneSeek Genome Profiler F-250 SNP chip (NEOGEN, 2017).

Several per-animal and per-marker quality control (QC) measures were applied in order to minimize bias in the process of identifying population structure and breed composition (Anderson et al., 2011). All QC steps were performed using the software

PLINK1.9 (Chang et al., 2015b). Per-animal QC measures included removal of samples with genotype completion rate less than 90%, and samples with pairwise IBS considered too high (> 0.98; S. Turner et al., 2011). Per-marker filters applied were: minor allele frequency (MAF) of less than 1%, genotype call rate of less than 90%, and HWE deviation with Chi-square P-value of less that $1x10^{-8}$ (Anderson et al., 2011). Markers in high LD were also pruned with window size of 5000 kilo base pair, step size of 10 base pair and LD threshold of 0.5 (Turner et al., 2011).

Identifying Breed Composition Using Whole Genome Data

Selecting unrelated animals

In both PCA and model-based analysis, efforts to identify population structure due to differences in breed composition can be biased by the presence of close familial relationships in the sample set being analyzed (Patterson et al., 2006; Alexander and Novembre, 2009; Matthew P. Conomos et al., 2016). Since it was known that there were certain animals with close familial relationship among animals included in the current study, it was important to account for the confounding effect of such relationships. To that end, a similar approach was used for both PCA (Matthew P Conomos et al., 2016) and model based analysis (Shringarpure et al., 2016) in which population structure identified in a subset of unrelated samples was used as a reference to characterize structure for the rest of the samples.

A subset of mutually unrelated animals that is representative of overall population structure was identified using an algorithm described by Conomos (2016) and implemented in the ‘pcairPartition’ function of the R package Genesis (Matthew P Conomos et al., 2016). This algorithm utilizes a pairwise kinship matrix estimated by the KING-robust method (Manichaikul et al., 2010) to identify a subset of mutually unrelated

samples. Unlike kinship estimation methods which assume a homogeneous population with no structure (e.g., IBD estimation implemented in PLINK ; Chang et al., 2015), the KING-robust method is not confounded by the presence of population structure (Manichaikul et al., 2010). Moreover, when applied to a set of samples with heterogeneous breed ancestry, the KING-robust method gives a systematically biased negative kinship estimate (termed divergence) for a given pair of unrelated samples with different breed of origin. This informative bias is used by the pcairPartition algorithm to include samples with divergent ancestry in the unrelated set so as to represent overall population structure (Matthew P. Conomos et al., 2016). Samples in the unrelated set are selected in such a way that they have pairwise kinship coefficient of less than 0.022 among them, whilst having the largest number of pairwise divergence of less than -0.022 with the rest of the samples (Matthew P Conomos et al., 2016). KING-robust pairwise kinship was calculated using the R function ’snpgdsIBDKing’ in the package Genesis (Matthew P. Conomos et al., 2016).

Model based analysis

Individual breed composition was estimated from genotype data using a maximum likelihood model implemented in the software ADMIXTUREv1.3 (Alexander et al., 2009; Shringarpure et al., 2016). ADMIXTURE uses genotype data to cluster individuals into subgroups, with the expected number of subgroups (termed *K*) specified beforehand. Subgroups memberships were taken as breed memberships, and pedigree information was used to identify the breed associated with a particular subgroup.

Using genotype data and a value for *K* as inputs, the model outputs two kinds of estimates stored in two matrices: *Q* and *F*. The number of columns in *Q* is equal to *K* whereas the number of its rows is equal to the number of samples included in the

analysis. Each column of *Q* contains membership coefficients of each sample to each subgroup. Since fractional subgroup membership is allowed, membership coefficients can also be conveniently interpreted as the proportion of an animals' genome contributed by a particular breed. The *F* matrix contains allele frequency estimates for the reference allele of each marker in each subgroup. The number of columns in *F* is equal to *K* whereas the number of rows is equal to the number of markers included in the analysis (Alexander and Novembre, 2009).

ADMIXTURE can be run in either supervised or unsupervised mode. In the unsupervised mode, both *Q* and *F* are estimated using genotype data and a *K* value as inputs. In the supervised mode, genotype data and a *K* value, along with an *F* matrix generated from a previous run on a reference population, are used as inputs to estimate *Q* (Alexander and Lange, 2015; Shringarpure et al., 2016).

The model used by ADMIXTURE assumes that SNP included in the analysis are in approximate LD and that all samples included in the analysis are mutually unrelated (Alexander and Novembre, 2009). The LD pruning step performed as part of QC is expected to minimize the effect of widespread LD in recently admixed populations such as the sample group included in this study (Alexander and Novembre, 2009). In order to control for the confounding effect of close familial relationships, the projection analysis feature of ADMIXTURE was used (Shringarpure et al., 2016).

To infer breed composition using the projection method of ADMIXTURE, a subset of unrelated samples was identified from the dataset as described earlier. ADMIXTURE was then run in the unsupervised mode on the unrelated subset, using genotype data and a *K* value of two as inputs, to obtain individual breed membership

coefficient (*Q*) and breed allele frequency (*F*) estimates. Genotype data for the remaining samples was then projected onto the population structure inferred for the unrelated samples (Shringarpure et al., 2016). In other words, breed allele frequencies (*F*) estimated for the unrelated set, along with genotype data for the rest of the samples and a *K* value of two, were utilized as an input when estimating breed membership coefficients (*Q*) for the rest of the samples using the supervised mode of ADMIXTURE (Shringarpure et al., 2016). A *K* value of two was chosen because it was known that all animals in the study derive their ancestry from are two parental breeds (Patterson et al., 2006; Zheng and Weir, 2016a).

Principal component analysis

Principal component analysis was applied to the genotype data using the 'pcair' function in the R package Genesis (Matthew P Conomos et al., 2016) to identify major axes of variation that explain most of the genetic structure in the study population (Patterson et al., 2006). The software minimizes confounding effect of close familial relationships in a manner similar to what is done in the projection analysis of ADMIXTURE – by identifying a set of unrelated samples, performing PCA on these, and then predicting PC values for the rest of the samples based on their genetic similarity to the unrelated set (Matthew P. Conomos et al., 2016). A subset of unrelated samples that are representative of overall population structure in the entire sample set were identified using the 'pcairPartition' algorithm as described earlier.

The 'pcair' algorithm first standardizes each column (i.e., SNP) of the genotype data for the unrelated set by subtracting the column mean and dividing by the column standard deviation. Genetic similarity matrix for the unrelated set was then obtained by multiplying the standardized genotype matrix by its transpose. Eigendecomposition was

then applied to the genetic similarity matrix to obtain eigenvectors, which correspond to PCs, and eigenvalues, which represent the variance of PCs. Principal Components for the rest of the samples, were then predicted by projecting their standardized genotype data onto the eigenvectors identified for the unrelated set (Matthew P. Conomos et al., 2016).

Results from PCA were compared to estimates of breed composition from ADMIXTURE and pedigree by visual representation of their relationship on a scatterplot and computing a Pearson's correlation coefficient.

Informative Marker Selection and Cross-validation

A small set of ancestrally informative markers that can be used to identify breed composition without a significant loss of accuracy were selected based on pairwise Fst (Wilkinson et al., 2011b). Representative allele frequencies for Angus and Brahman were calculated after identifying two sample groups with ancestry coefficient of more than 0.9 for the respective breeds as estimated by ADMIXTURE. Fixation index of each SNP in the full genotype data (post QC) was then estimated using Weir and Cockerham's method (Weir and Cockerham, 1984) implemented in PLINK.

After ranking based on Fst, 60 subsets of SNP were selected, starting from the top five SNPs and increasing the number by five up to 300 SNPs. The ability of the markers in each of these subsets to predict breed composition (Angus proportion) was evaluated by means of a five-fold cross-validation scheme. The sample dataset was randomly divided into five groups or folds. For the first round of cross validation, four folds were used as a training set to estimate parameters (β) of a linear regression model (Equation 1) in which the dependent variable was the Angus proportion estimated by ADMIXTURE using full genome SNP data, and the independent variables were genotype values for

selected SNP. Model parameters (β) estimated for the training set were then used to predict Angus proportion in the fifth group or fold which was used as a validation set (Equation 2). The same set of SNPs that were used in the model-training step (Equation 1) were also used for the predictive model in the validation step (Equation 2). Accuracy of prediction made for animals in the validation set using a given set of Fst selected SNPs was measured as Pearson's correlation coefficient with Angus proportion estimated by ADMIXTURE run on full genome SNP data. Four more rounds of cross validation were carried out by rotating the folds until all five were used as both training and validation set. Therefore, five correlation coefficient values were produced by each five-fold cross validation routine applied to a given set of selected SNPs. In addition, for all 60 sets of SNPs selected based on Fst, the cross-validation process described above was replicated five times, with the random five-way partitioning of the sample dataset taking place during each replication round. Consequently, for each set of selected SNPs, 25 correlation values were produced and summarized. For the purpose of comparison, the replicated cross validation procedure was repeated with the only difference this time being that SNP selection was random instead Fst-based. All cross validation steps were carried out using scripts written in the R programming language (R Core Team, 2016) which can be found in the Appendix section.

$$y_j = \beta_0 + \beta_1 * SNP_1 + \beta_2 * SNP_2 \ \ldots + \beta_i * SNP_i + \varepsilon_i \quad (1\text{-}1)$$

For i selected SNP and individual j

$$\hat{y}_i = \beta_0 + \beta_1 * SNP_1 + \beta_2 * SNP_2 \ \ldots + \beta_i * SNP_i \quad (1\text{-}2)$$

For i selected SNP and individual j

Results and Discussion

Genotype Data Quality Control

Six hundred and seventy-six animals were kept after removing 104 with a genotype completion rate of less than 90% and a pair of samples with IBS of 0.998. From an initial set of 221,077 SNP, a subset of 54,728 SNP was kept after removing 64,496 SNP for low MAF (< 1 %) and 48,386 SNP for failing to meet minimum call rate; 8,088 SNP for Hardy-Weinberg Equilibrium deviation and 45,379 SNP due to the LD pruning step. Therefore, a total of 54,728 SNP and 676 cattle passed QC, and they were used in subsequent analysis.

Identifying a Subset of Unrelated Samples

The R function ‘pcairPartition’ identified 74 samples as unrelated and ancestrally representative of the entire sample set as compared to the rest of the samples. This partitioning was based on pairwise kinship and divergence estimates by the King-robust method (Matthew P. Conomos et al., 2016). The distribution of king-robust estimated for all pairwise comparisons (n= 228,150) is shown in Figure 1-1. It can be seen in this figure that the majority of the estimates were negative, indicating a highly heterogeneous ancestral background of the samples (Matthew P. Conomos et al., 2016).

In such a population, methods for estimating kinship using genetic data which do not account for the presence of population structure (e.g., IBD estimation model in PLINK) will tend to overestimate kinship between related animals with the similar breed background while underestimating kinship between related animals with different breed background (Thornton et al., 2014). In contrast, the king-robust method is robust to the presence of population structure, but it will give negatively biased estimates for a pair of

unrelated samples with divergent breed ancestry (Matthew P. Conomos et al., 2016). If the interest was only in kinship estimation, all the negative estimates would be truncated to 0 (Weir and Goudet, 2016). However, in the current analysis, these negative estimates were considered as measures of divergence (Matthew P Conomos et al., 2016). As described in the Materials and Methods section, they were used to identify samples with the most divergent breed background, and hence the most representation of population structure due to breed composition difference in the entire sample set (Thornton et al., 2014; Matthew P. Conomos et al., 2016).

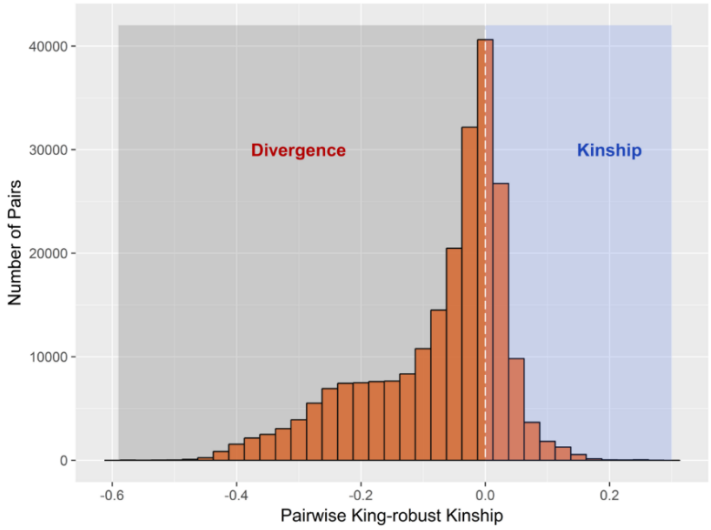


Figure 1-1. Histogram showing the distribution of pairwise kinship and divergence estimates by King-robust method among all samples.

Model Based Analysis

As expected (Matthew P Conomos et al., 2016), the unrelated set of animals identified by pcairPartition, was representative of the overall population structure, and contained all animals with zero or one breed membership coefficients for both breeds. This is illustrated in Figure 1-2; a bar plot of the *Q* matrix from an unsupervised ADMIXTURE run on this set. Figure 1-3 shows another bar plot of the *Q* matrix from a

supervised ADMIXTURE run on the related set of samples where the *F* matrix estimated for the unrelated set was used as an input. This is expected to minimize the confounding effect of close familial relationships on breed membership inference for the related set of samples.

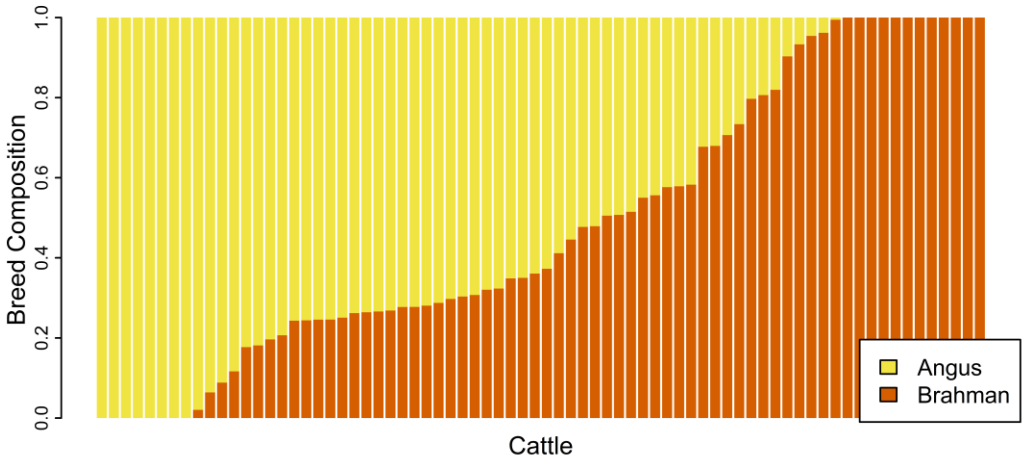


Figure 1-2. Bar plot showing the proportion of the genome contributed by each breed for 74 unrelated samples. The proportions were obtained from a model based estimation (ADMIXTURE1.3) of breed composition using genomic data only. We can see here that this group is ancestrally representative of the entire sample set.

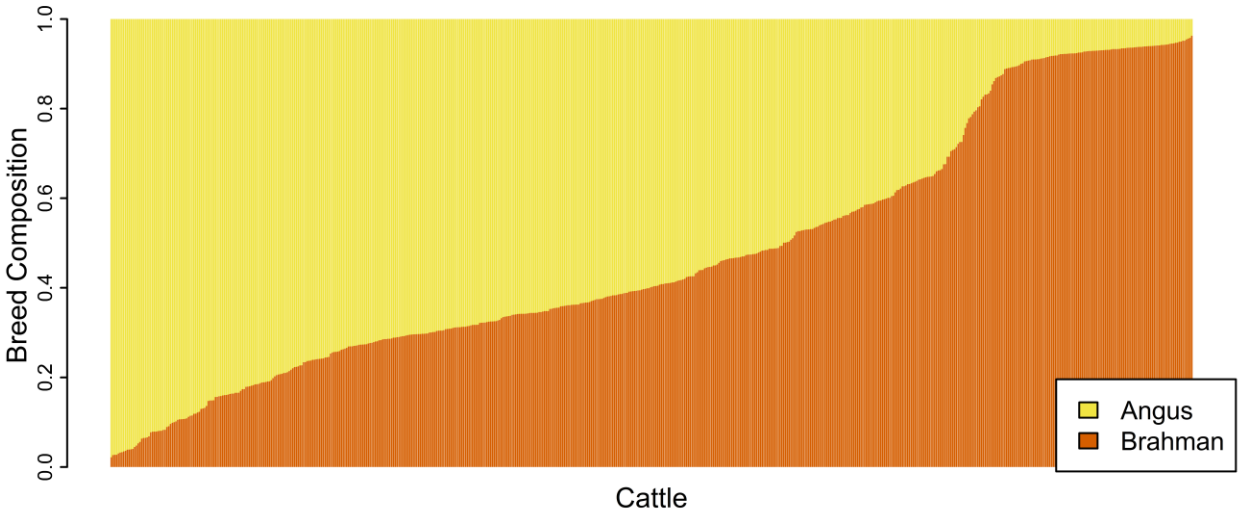


Figure 1-3. Bar plot showing the proportion of the genome contributed by each breed for 602 samples in the related set as obtained from a supervised ADMIXTURE run using the unrelated set as a reference population.

There was a very strong correlation (R=0.965) between breed composition estimates for either breeds from ADMIXTURE and the estimates obtained using pedigree records. This result is in agreement with other studies in which pedigree-based breed composition estimates were compared with estimates using genome-wide SNP data. In one such study, Frkonja et al (2012b) compared different methods of estimating breed composition using a set of 495 bulls consisting of purebred Red Holstein Friesian, purebred Simmental, and their crossbreds. This study reported a correlation coefficient of 0.972 between breed proportions obtained from pedigree and breed membership coefficients estimated by STRUCTURE using 40,492 genome-wide SNPs. Another similar study (Dodds et al., 2014) found a correlation coefficient of 0.89 between breed composition estimates from pedigree and from STRUCTURE run on a set of 10,000 SNP for a total of 4,944 sheep consisting of four different breeds of sheep and their crossbreds.

However, similar to the other studies, there were discrepancies between breed composition estimates from genome-wide data and pedigree for certain samples (Figure 1-4). The mean and standard deviation of the absolute difference between breed composition estimates from the two methods were 0.056 and 0.060, respectively. For 72% of the animals, the difference was within 1 standard deviation, and 5 % had a difference of more than two standard deviations.

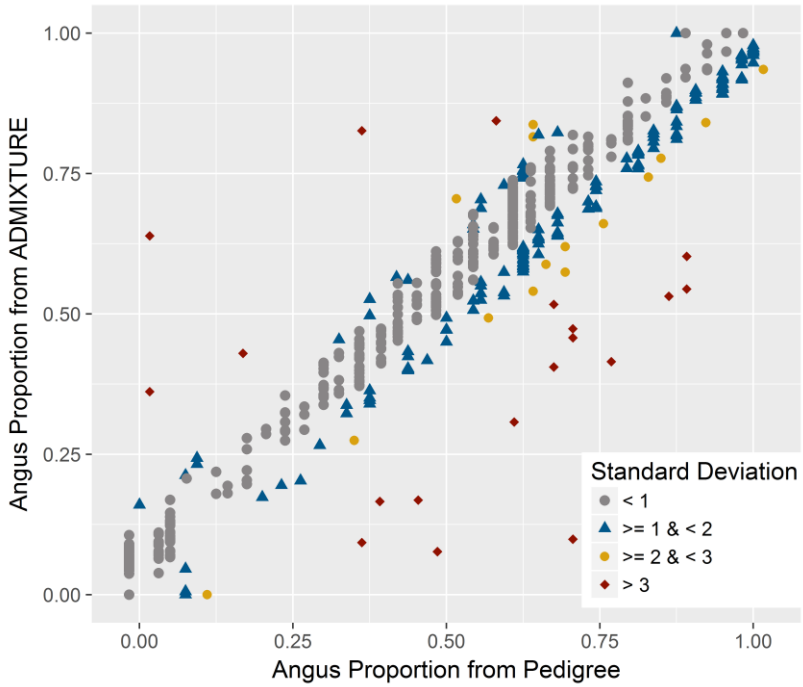


Figure 1-4. A scatter plot showing a strong positive relationship between Angus ancestry estimates from ADMIXTURE and Pedigree. However, for certain samples, there was discrepancy between the two estimates. The color for each animal corresponds to the amount of standard deviation by which the two measures differ.

For crossbred animals, breed composition derived from genomic data is more accurate than pedigree-based estimates since pedigrees can be incomplete or incorrect (Frkonja et al., 2010; Vanraden and Cooper, 2015). Mendelian sampling during recombination can also lead to deviation from composition expected based on pedigree (L. A. Kuehn et al., 2011). On the other hand, estimates based on genomic data can also be biased or loose accuracy under certain conditions. One factor that can lead to inaccurate estimates is sample selection bias which can be described as failure to include sufficient samples that are representative of all parental breeds in the analysis (Long, 1991; Shringarpure and Xing, 2014). Weak differentiation between parental breeds can also lead to lower accuracy when estimating proportional contribution from these breeds in crossbred animals using genetic data only, (Patterson et al., 2006; L. A.

Kuehn et al., 2011). An example of breeds that can prove difficult to differentiate due to weak differentiation are Angus and Red Angus (L. A. Kuehn et al., 2011).

Principal Component Analysis

The first and second PCs explained 27% and 5.6% of the variation in the entire genetic data, respectively. The fact that PC1 explained much more variation in the genetics data as compared to the rest of the PCs is consistent with there being two major parental breeds (Patterson et al., 2006; McVean, 2009). McVean (McVean, 2009) mentioned that the proportion of variation explained by PC1 in a two-way admixture is actually closely related to the Fst estimated between the two parental populations, which makes intuitive sense since Fst is the ratio of between-population variation to overall variation. The genome-wide Fst average (0.25) found in the current study was close to the proportion of variation explained by PC1 (0.27), consistent with McVean's (2009) observation of the relationship between the two. The slightly lower Fst in the current study (as compared to PC1) could be explained by the fact that the 'purebreds' used for Fst calculation actually consisted of animals having more than 90% of either breed, not 100%.

The first PC had a very strong correlation (R=0.966) with breed composition derived from pedigree. The relationship between PC1 and pedigree-based breed composition is also illustrated in Figure 1-5, which shows that position along PC1 corresponds with breed composition. However, it can also be seen in the same figure that the position of certain animals along PC1 is not consistent with their pedigree information. Such inconsistencies, similar to discrepancies between pedigree-derived breed composition and the result from ADMIXTURE, are likely due to inaccuracies in

pedigree records and/or the effect of Mendelian sampling (Patterson et al., 2006; L. A. Kuehn et al., 2011; Vanraden and Cooper, 2015).

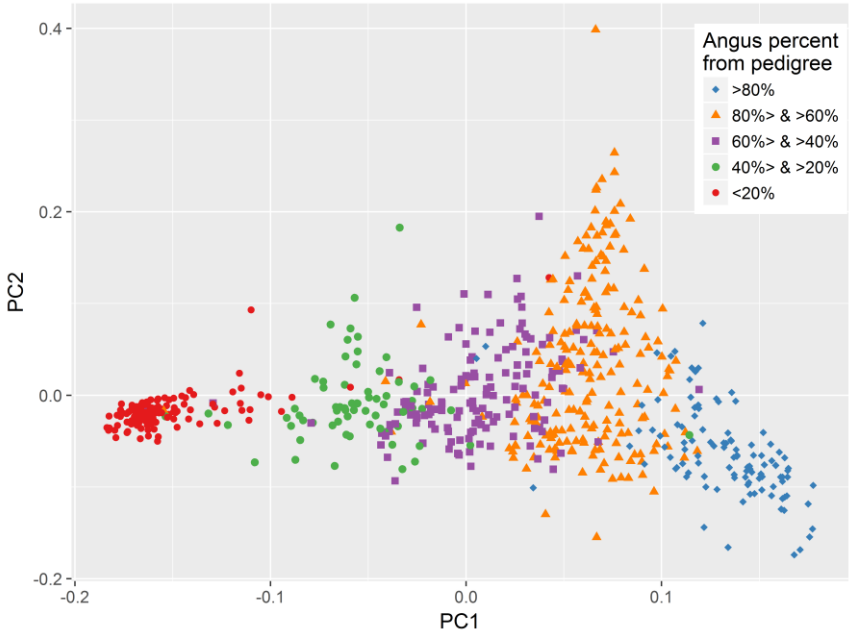


Figure 1-5. A PC1 versus PC2 scatter plot showing how the first PC agrees with pedigree information. However, it can also be seen that the position of certain samples along PC1 is not consistent with what would be expected from pedigree records.

It can also be seen in Figure 1-5 that animals with around 2/3 Angus proportion had more scatter along PC2. The cattle coming from least one generation of Brangus-Brangus mating had the most scatter along PC2, and were largely not located along a line connecting the two clusters formed by the purebreds (Figure 1-6). In contrast, F1 and first generation Brangus cattle showed much less variation across PC2, and were located along the line connecting the two clusters formed by the purebreds (Figure 1-6), as would be expected in the case of a recent two-way admixture (Patterson et al., 2006; McVean, 2009). The distinct pattern of variation seen in the cattle born from Brangus-Brangus matings is likely due to the extended number of generations since the initial crossing of the parental breeds in these animals (Patterson et al., 2006). Close familial

relationships can be ruled out as a cause since such pattern was also seen in the unrelated set of samples (Figure 1-7). In addition, it has been demonstrated (McVean, 2009) that, in populations resulting from a two-way admixture, the proportion of genetic variation explained by the first PC drops as the number of generations since the initial admixture event increases. This could explain why animals from at least one generation of Brangus-Brangus mating have little variation across PC1 as compared to PC2.

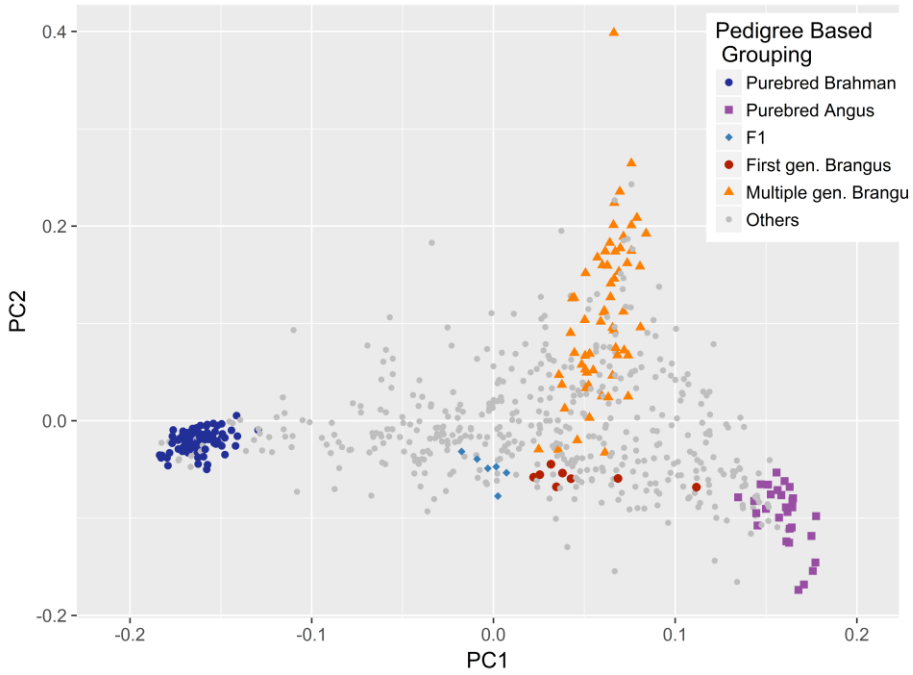


Figure 1-6. When plotting the first PC against the second, Brangus cattle coming from at least one generation of Brangus-Brangus mating had more scatter across PC2. In contrast, first generation Brangus and F1 cattle showed minimal scatter across PC2 and were positioned along the line connecting the two clusters made by the purebreds.

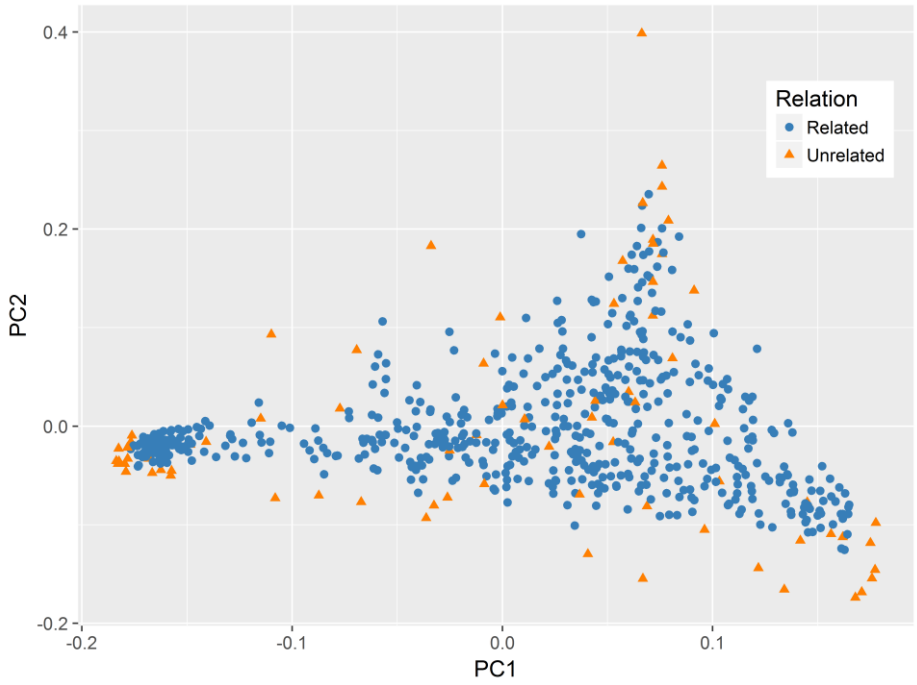


Figure 1-7. PCA was performed on the unrelated set of animals (blue) and PC1 & 2 values for the rest of animals (orange) were predicted based on their genetic similarity to animals in the unrelated set.

The first PC had a very strong correlation (R=0.999) with Angus/Brahman proportion from ADMIXTURE. A similar result was reported in a previous study (Patterson et al., 2006) in which a correlation coefficient of 0.995 was obtained between PC1 and model estimates for European ancestry in an admixed human population. Despite apparent differences in their approach, both PCA and model-based methods are closely related and can be viewed as different ways of factorizing the genotype matrix (Engelhardt and Stephens, 2010a). While the first PC from PCA is sufficient to measure the level of admixture in a crossbred population with two parental breeds, model-based methods need two coefficients of membership for both breeds to provide the same information (Patterson et al., 2006; Engelhardt and Stephens, 2010a).

Notwithstanding the strong correlation between PC1 and breed membership coefficients estimated by ADMIXTURE, the relationship between the two appeared to

be different for the related and unrelated set of samples as illustrated in Figure 1-8. For the related set, there was a linear relationship between the two values, whereas for the unrelated set, ADMIXTURE estimates had more extreme values at both ends. A similar observation was made by Engelhardt and Stephens (2010a) who noticed that, when applied to an admixed set of samples with divergent ancestral groups, ADMIXTURE tends to give cluster membership estimates that are more extreme as compared to components from PCA. According to Engelhardt and Stephens (2010a), this tendency has to do mainly with differences in the type of constrains imposed during optimization when estimating the *Q* matrix and PCs in ADMIXTURE and PCA, respectively. Another contributing factor could be the difference in the assumed distribution of the errors associated with estimates in the two methods.

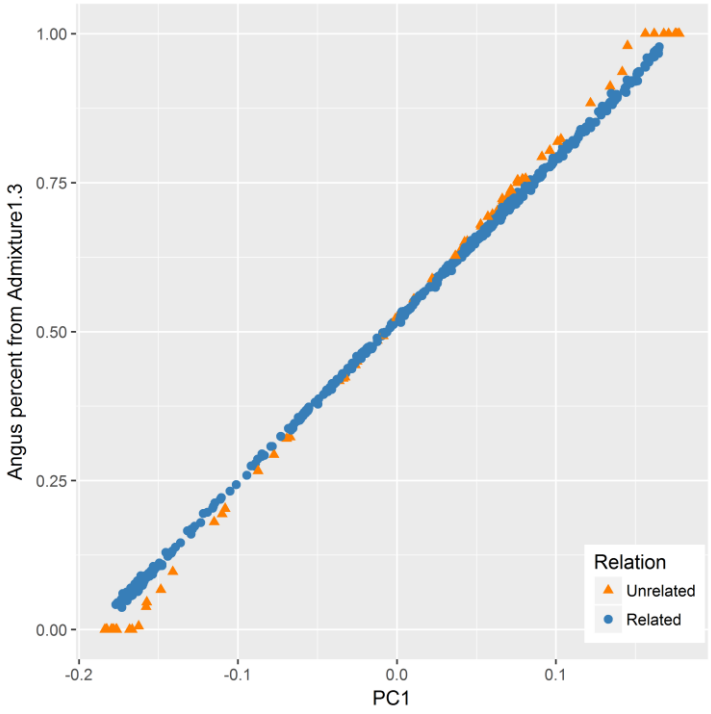


Figure 1-8. The plot shows the relationship between PC1 and Angus percent estimated by ADMIXTURE for both related and unrelated set of samples. Angus proportion from ADMIXTURE tended to be more extreme as compared to PC1 values for the unrelated set.

Because of the need to explain overall genetic variation or population structure in terms of ancestry from a predefined number breeds (*K*), ADMIXTURE estimates membership coefficients to all *K* breeds using a constrained optimization process (via quadratic programming) which forces the coefficients to be non-negative and to sum to one (Alexander et al., 2009). This means that overall genetic variation is represented only by *K* number of variables, which correspond to membership coefficients to the respective breeds, without attributing variation to any other source. In contrast, PCA does not impose such constraints, in that a predefined number of PCs are not forced to explain all of the genetic variation (Engelhardt and Stephens, 2010b). Instead, when applied to data with n number of genetic variants, PCA estimates n PC, each explaining certain proportion of the overall genetic variation. In a two-way admixture, as is the case in the current study, PC1 captures genetic variation due to heterogeneous breed ancestry (Patterson et al., 2006). However, variations due to additional factors such as familial relationships are also captured by subsequent PC (McVean, 2009; Engelhardt and Stephens, 2010b). Consequently, PC1 values tend to be less biased towards either end of the admixture spectrum as compared to membership coefficient estimates by ADMIXTURE for both ancestral groups (Engelhardt and Stephens, 2010b). Another factor contributing to the relatively extreme nature of ADMIXTURE estimates could be its assumption that the errors have a binomial distribution whereas PCA assumes the errors have a Gaussian distribution (Engelhardt and Stephens, 2010b).

As compared to ADMIXTURE, PCA is appealing in that it is computationally more efficient while providing a similar level of information as model-based clustering (Patterson et al., 2006). Furthermore, visual representations based on the top few PC

provide better insights into the diversity and extent of demographic events underlying different levels of population structure (McVean, 2009). Nonetheless, the main issue with PCA is interpretability (Patterson et al., 2006). For instance, in the current study, cluster membership coefficients estimated by ADMIXTURE were interpreted as Angus and Brahman proportions. In contrast, there was no such interpretation for PC1 values, which ranged from -0.18 to 0.18. Although there have been suggestions (Patterson et al., 2006; McVean, 2009; Zheng and Weir, 2016b) on how to interpret PCA results in terms of admixture levels or genealogy, caution should be taken when doing so since different demographic events result in similar PCA projections (McVean, 2009).

Informative Marker Selection and Cross-Validation

A small number of informative SNPs selected by Fst were able to predict Angus percent with high accuracy as measured by Pearson's correlation coefficient with Angus ancestry estimated by ADMIXTURE using full genome data (Figure 1-9). As few as five SNP were sufficient to predict Angus percent with an accuracy of ~0.9, whereas an accuracy of ~0.95 was reached with 15 SNP. A plateau of ~0.99 was reached with 90 SNP. In comparison, five randomly selected SNP had an accuracy of ~0.6, and there was not much improvement after 135 randomly selected SNP at which an accuracy of ~0.96 was obtained.

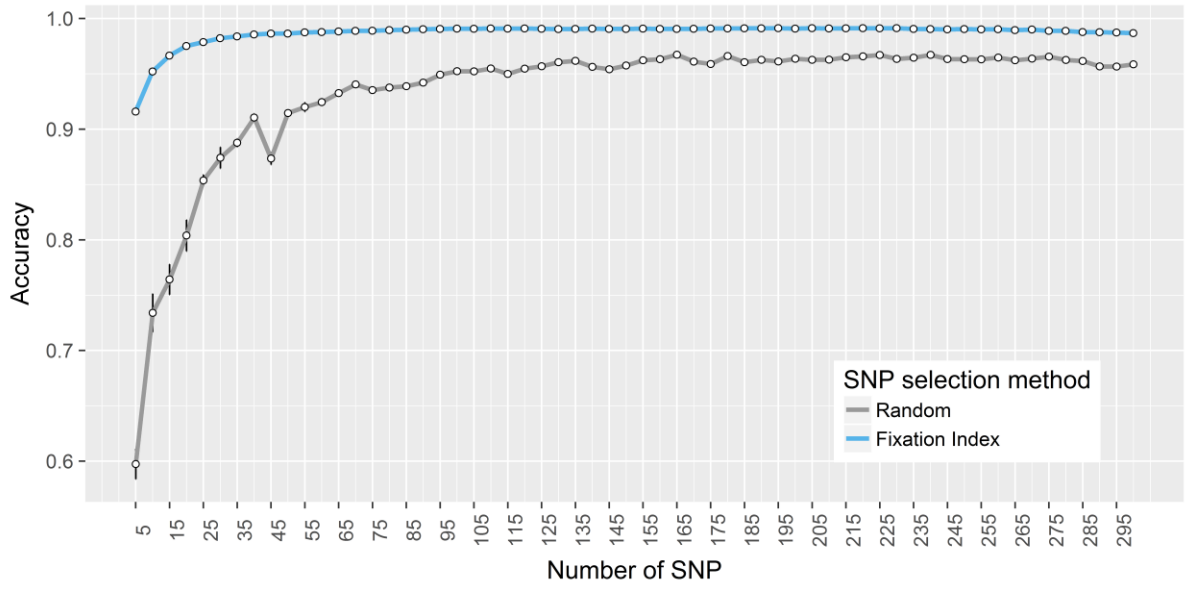


Figure 1-9. Each point represents the mean of 25 accuracy values from a 5-fold cross validation replicated 5 times performed on a single set of selected SNPs. The bars around the points represent standard errors. Accuracy of prediction was measured as correlation with Angus

Breed composition was predicted with higher accuracy using fewer markers in the current study as compared to previous similar other studies. Frkonja et al (2011) reported that 48 SNP, which were selected from 40, 492 SNP based on Fst values, predicted breed composition with an accuracy of 0.9 in cattle that are crossbreds between Simmental and Red Holstein Frisian. This study used pedigree as a reference when calculating accuracy of prediction. Another comparable study by Wilkinson et al (2011a) used 60 SNP selected from 40,483 SNP based on pairwise Fst to assign 384 purebred cattle sampled from 17 breeds to their respective breeds with an accuracy of 0.95. However, in contrast to the current study, Wilkinson et al (2011a) did not include crossbreds and attempt to estimate admixture levels.

The high level of prediction accuracy with minimal number of markers observed in the current study can be attributed to two major factors. One has to do with the level

of differentiation between the two breeds. It has been previously noted (Maudet et al., 2002; Patterson et al., 2006; McVean, 2009; L A Kuehn et al., 2011; Wilkinson et al., 2011a) that the number of SNP required to differentiate between a given pair of breeds is inversely proportional to the amount of differentiation them. Genome-wide average Fst between Angus and Brahman purebreds in the current study (0.25) was much higher than between Simental and Royal Holstein Frisian (0.11) in (FRKONJA et al., 2011) , which explains why fewer SNP performed better in the current study. Another factor is that only two breeds and their crosses were involved. The number of markers needed to properly differentiate between different breeds increases with the number of breeds involved (L. A. Kuehn et al., 2011; Lewis et al., 2011)

It was interesting to note that, although not as good as SNPs selected based on Fst, a small number of randomly selected SNPs contained sufficient information to allow prediction of breed percent with high accuracy. Similar to Fst selected SNPs, the breed informativeness of a small number of randomly selected SNPs can be linked to the level of differentiation between the two breeds (Patterson et al., 2006; McVean, 2009). Another likely contributing factor is widespread LD associated with recent admixture between two differentiated populations, which is the case in the current study population (Patterson et al., 2006). Extensive LD will make it more likely for a large number of non-informative SNP to be on the same linkage block as informative markers. Although non-informative SNP on such linkage blocks appear to provide information about breed composition, their informativeness will drop as LD breaks down with increasing number of generations since initial admixture (Patterson et al., 2006).

An additional contributing factor to the informativeness of randomly selected SNPs could be the ascertainment process when constructing the SNP panel used in the current study (GeenSeek GGPF250) which included predominantly taurine breeds as compared to zebu (Schnabel et al., 2016). This most likely led to MAF for most SNP in the panel being higher in Angus than in Brahman breeds, making it likely for randomly selected SNP to differ in frequency between the two breeds (Lachance and Tishkoff, 2013). Figure 1-10 illustrates how mean MAF for 54,728 SNP used in the current study (post QC) differ in 10 sets of samples grouped based on Angus-Brahman percent as estimated by ADMIXTURE. It can be seen that there is slight but consistent decrease in average MAF as the amount of Brahman percent increases, which supports the suggestion that ascertainment bias may have contributed to the breed informativeness of randomly selected SNP.

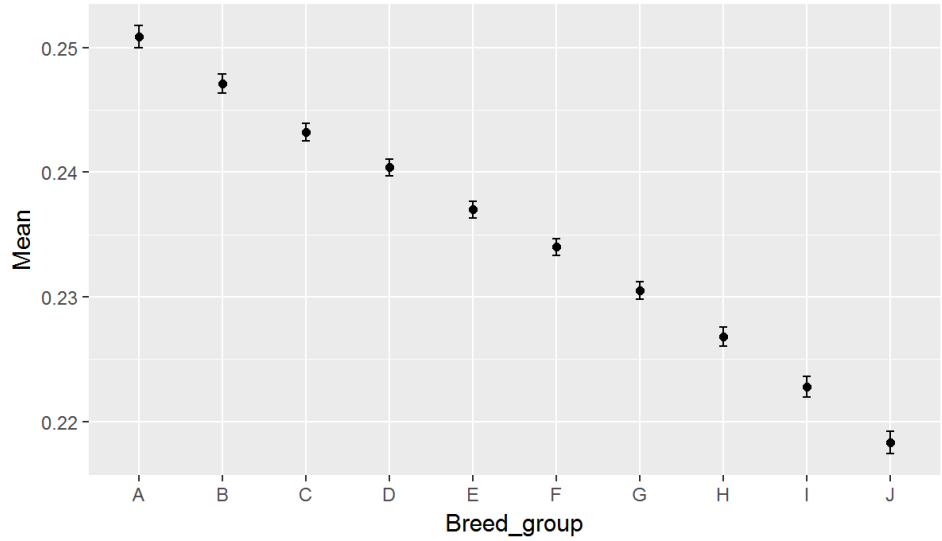


Figure 1-10. The plot illustrates the drop in the genome-wide average MAF as the Angus percent decreases from >90% (A) to <10% (J).

Conclusion

By applying PCA and the maximum likelihood method of ADMIXTURE to genomic data, it was possible to successfully characterize population structure resulting from heterogeneous breed ancestry, while accounting for close familial relationships. Principal component analysis results offered better insight into the different hierarchies of genetic variation structuring. While PC1 was strongly correlated with Angus-Brahman proportions, PC2 represented variation within animals that have a relatively more extended Brangus lineage – indicating the presence of a distinct pattern of genetic variation in these cattle.

In contrast, ADMIXTURE estimates of breed composition forced all genetic variation to be explained only in terms of Angus and Brahman proportion represented by columns of the *Q* matrix (Figures 1-2 & 1-3), without accounting for other sources of variation. The effect of such a constraint was that, for the unrelated set, ADMIXTURE estimates tended to be close to either zero or one (i.e., purebreds of either breeds), as compared to PC1 (Figure 1-8).

On the other hand, in the related set, ADMIXTURE estimates had very good agreement with PC1 and did not have bias towards either zero or one as compared to those for the unrelated set (Figure 1-8). This is likely due to measures taken to account for sources of genetic variation other than breed ancestry (e.g., familial relationships) by using the unrelated set as a reference population. This shows how breed composition inferences made by ADMIXTURE-like methods (e.g., STRUCTURE, fastSTRUCTURE and Frappe) can be confounded by other sources of population structure and highlights the importance of accounting for such sources by using an unrelated, breed-representative reference population.

Although there was strong agreement between breed proportions estimated from pedigree and genetic information, there were significant discrepancies between these two methods for certain animals (Figure 1-4 & 1-5). This is most likely due to inaccuracies in pedigree information of these animals, which supports the case for using genomic information to complement and/or replace pedigree information when estimating breed composition.

Using a small subset of SNP, which were selected based on pairwise Fst between representative samples from the two breeds, it was possible to predict breed composition with high accuracy. This result will be a valuable input for the development of a SNP panel for identifying breed composition in Angus-Brahman crossbreds with minimal cost. Such a panel can help improve the efficiency of Angus-Brahman crossbreeding programs. An additional use could be cheap independent authentication of breed of origin in breed-labeled beef products.

CHAPTER 2
GENOMICS OF RESILIENCE TO CLIMATIC STRESSORS IN SHEEP

Introduction

Climate change along with population growth and increased food demand are expected to place a significant amount of strain on livestock production in the not too distant future (Thornton et al., 2009; Boettcher et al., 2014). To be able to cope with these changes, the livestock industry has to improve in terms of both efficiency and productivity whilst being less prone to harsh environmental conditions (Boettcher et al., 2014). To achieve such improvements, substantial changes have to be made to different aspects of animal production and husbandry. These include housing, nutrition, health and genetics (Boettcher et al., 2014).

Genetic diversity is positively correlated with the adaptive potential of an organism (Hoffmann et al., 2015; Matuszewski et al., 2015; Ellegren and Ellegren, 2016). Relatedly, having diverse genetics in livestock will allow selection of available stock or development of new breeds in response to a wide range of conditions including climate change and variability (Hoffmann, 2010). Unfortunately, although modern animal breeding has led to a steep growth in productivity, it has also increased vulnerability of livestock to adverse environmental conditions, mainly through degradation of genetic diversity (Groeneveld et al., 2010). Selection focused mainly on production traits with little consideration to adaptability traits such as disease resistance and thermal tolerance has led to a reduction in genetic diversity (Drucker et al., 2001; FAO, 2007). Additionally, in certain livestock species (e.g., dairy cattle), the use of a small number of highly productive males to inseminate a large number of females has led to a reduction in the effective population size (Drucker et al., 2001; Kijas et al., 2012). Therefore,

maintaining existing genetic diversity and finding an optimum balance between productivity and adaptability are some of the challenges that need to be addressed in animal breeding (Nardone et al., 2010).

Understanding the genetic background of environmental adaptation is an important step towards incorporating adaptability traits into genomic breeding programs (Hayes et al., 2009). The goal of the current study was to use Environmental Association (EA) analysis to characterize the genomic background of environmental adaptation in sheep sampled from parts of the US with divergent climatic condition. The specific objectives of the study were to:

1) Identify loci showing adaptive variation while accounting for background neutral genetic variation and
2) Identify biological processes affected by environmental variables by performing gene ontology term enrichment using candidate genes identified by EA analysis.

Literature Review

Climate, Climate Variability and Climate Change

A glossary by the Intergovernmental Panel for Climate Change (IPCC) gives both narrow and broad sense definition for climate. In the narrow sense, climate is defined as the average weather (involving variables such as temperature, precipitation and wind), usually over a 30 year period (Barros et al., 2012). In the broad sense, climate is defined as the state of the climate system, which is a highly complex system consisting of five major components: the atmosphere, the oceans, the cryosphere, the land surface, the biosphere, and the interactions between them (Barros et al., 2012).

Climate variability is a term used to describe changes in the climate system on time scales of a few years to a few decades (i.e., shorter than a climatic change

averaging period). Climate variability is attributed to Pacific Oscillation (IPO), which is responsible for decadal scale variability in the Pacific basin, El Niño/Southern Oscillation (ENSO), which causes inter-annual variability throughout many tropical and subtropical regions and the North Atlantic Oscillation (NAO), which causes climate perturbations over Europe and northern Africa. Global warming is known to have a significant effect on IPO, ENSO and NAO as well (Collins et al., 2010).

Climate change refers to any considerable change in Earth's climate that lasts for an extended period of time, typically decades or longer (Barros et al., 2012). An increase in the average temperature of the lower atmosphere due to factors associated with climate change is referred to as global warming (Pachauri and Meyer, 2014). Climate change is caused by a combination of natural and anthropogenic factors, the effect of which is measured in terms of radiative forcing (RF) which is defined as the difference between the solar energy absorbed by the Earth and the energy radiated back to space. Radiative forcing is measured in watts per square meter (W/m^2) measured at the tropopause. A factor (also known as forcing) with positive RF leads to near-surface warming whereas one with negative value leads to cooling (Pachauri and Meyer, 2014).

Forcings can be natural or anthropogenic. Natural forcings include solar irradiance and volcanic aerosols. Volcanic aerosols can have a largely cooling effect on the climate system for some years after major volcanic eruptions, whereas changes in total solar irradiance are thought to have contributed only around 2% of the total radiative forcing in 2011 relative to the beginning of the industrial revolution (1750AD; Pachauri and Meyer, 2014).

The main forms of anthropogenic forcings are greenhouse gases (GHGs) such as carbon dioxide (CO_2), methane (CH_4) and nitrous oxide (N_2O), which lead to positive RF (Pachauri and Meyer, 2014). Atmospheric levels for these gases is the highest it has been in at least 800,000 years (Pachauri and Meyer, 2014). The concentration of GHGs has increased considerably since the start of the industrial revolution (1750; CO_2 by 40%, CH_4 by 150% and N_2O by 20%). The total anthropogenic RF over 1750–2011 was 2.3 (1.1 to 3.3) W/m^2 (Forster et al., 2007; Pachauri and Meyer, 2014), and CO_2 was the largest single contributor. About half of the cumulative anthropogenic CO_2 emissions between 1750 and 2011 have occurred in the last 40 years (Pachauri and Meyer, 2014). Multiple studies have concluded that anthropogenic forcings, a major component of which is CO_2, are responsible for the majority of the observed increase in global average surface temperature (Forster et al., 2007; Pachauri and Meyer, 2014).

Projections for Future Climate

The effect of global warming, superimposed on that of inter-annual and inter-decadal variabilities, is expected to have a considerable effect on the climate system (Salinger et al., 2005). These include a decrease in cold temperature extremes, an increase in warm temperature extremes, an increase in extreme high sea levels and an increase in the number of heavy precipitation events in a number of regions (Salinger et al., 2005; Pachauri and Meyer, 2014).

The main factor that is expected to dictate global mean surface warming by the late 21st century is the aggregated amount of CO_2 emissions (Pachauri and Meyer, 2014). Projections of GHG emissions depend on both socio-economic development and adoption of policies affecting response to climate change (Pachauri and Meyer, 2014). Climate researchers have outlined 4 possible scenarios that could take place depending

on the extent to which emissions have been mitigated (Pachauri and Meyer, 2014). The scenarios are known as Representative Concentration Pathways (RCPs) and are based on published studies on GHG emission scenarios (Meinshausen et al., 2011). The 4 RCPs include a stringent mitigation scenario (RCP2.6), two intermediate scenarios (RCP4.5 and RCP6.0) and one scenario with very high GHG emissions (RCP8.5). The numbers next to the RCP represent the RF in W/m^2 associated with the specific scenario (Moss et al., 2010). Scenarios without additional efforts to constrain emissions (’baseline scenarios’) lead to pathways ranging between RCP6.0 and RCP8.5. A stringent mitigation scenario (RCP2.6) is expected to keep global warming within 2°C of pre-industrial temperatures (Meinshausen et al., 2011; Pachauri and Meyer, 2014).

The increase in global mean surface temperature by the end of the 21st century (2081–2100), relative to 1986–2005, is likely to be 0.3°C to 1.7°C under RCP2.6, 1.1°C to 2.6°C under RCP4.5, 1.4°C to 3.1°C under RCP6.0 and 2.6°C to 4.8°C under RCP8.5 (Pachauri and Meyer, 2014). Changes in precipitation will not be uniform. The high latitudes and the equatorial Pacific are likely to experience an increase in annual mean precipitation under the RCP8.5 scenario (Pachauri and Meyer, 2014). In many mid-latitude and subtropical dry regions, mean precipitation will likely decrease, while in many mid-latitude wet regions, mean precipitation will likely increase under the RCP8.5 scenario (Pachauri and Meyer, 2014). Extreme precipitation events over most of the mid-latitude land masses and over wet tropical regions will very likely become more intense and more frequent (Pachauri and Meyer, 2014).

Impacts of Climate Change on Food Security

In temperate regions, higher temperatures are expected to be mostly beneficiary to agriculture in terms of expansion of areas potentially suitable for cropping, and rise in

crop yields due to increase in length of the growing period (Reilly et al., 2003; Parry et al., 2004; Schmidhuber and Tubiello, 2007). However, an increased frequency of extreme events such heat waves and droughts in the Mediterranean region or heavy precipitation events and flooding in temperate regions may negate the gains mentioned above (Schmidhuber and Tubiello, 2007; Ebi and Bowen, 2016). Climate change above 3°C will risk overall decreases in the global food production capacity that might have a heavy impact even in places where food production remains adequate locally (Beddington et al., 2012). Together with the inevitable increase in food demand, a global temperature increase of ~4°C or more above late 20th century levels would threaten food security throughout the world (Pachauri and Meyer, 2014).

Impact of Climate Change on Livestock Production

Climate change and variability affect livestock production directly through, for example, heat stress, and indirectly through effect on forage quality and distribution of livestock diseases (Backlund et al., 2008). In the event of exposure to high ambient temperature, physiological and metabolic adjustments resulting from thermoregulatory responses to thermal stress have negative consequences on animal productivity and health, mostly through reduced feed intake aimed at minimizing heat production from consumption and metabolic utilization of feed (Nardone et al., 2010; Renaudeau et al., 2012).

Exposures to sudden and extreme weather without sufficient time for conditioning have resulted in considerable losses in the domestic livestock industry in the US. Some feedlots lost more than 100 head each during severe heat wave episodes in 1992, 1995, 1997, 1999, 2005, and 2006 (Backlund et al., 2008). The heat waves in 1999 were particularly severe in Nebraska where a loss of 5,000 heads was recorded

(Nienaber et al., 2007). In 2006 a major heat wave moving across the USA resulted in the death of 25,000 cattle and 700,000 poultry in California (Renaudeau et al., 2012). Moreover, economic losses from reduced cattle performance due to these extreme conditions were likely several times greater than losses from cattle deaths (Backlund et al., 2008). Across the US, heat stress results in estimated total annual economic losses to livestock industries that are between $1.69 and $2.36 billion in 2003 (St-Pierre et al., 2003). Of these losses, $897 to $1500 million occur in the dairy industry, $370 million in the beef industry, $299 to $316 million in the swine industry, and $128 to $165 million in the poultry industry (St-Pierre et al., 2003).

Livestock production is also influenced by climate change and variability indirectly through changes in pasture quality/quantity and disease distribution (Backlund et al., 2008; van Dijk et al., 2010). Elevated atmospheric CO^2 can increase the carbon to nitrogen ratio in forages and thus reduce the nutritional value of those grasses, which in turn affects animal weight and performance (McNeill, 2010). Under elevated CO_2, a decrease of C_4 grasses and an increase of C_3 grasses may occur, which could potentially reduce or alter the nutritional quality of the forage grasses available to grazing livestock (McNeill, 2010). In addition, it has been reported that climate change induced encroachment of woody plants into grasslands has had a sizable negative impact on the range livestock industry (Backlund et al., 2008).

Shifts in temperature and precipitation patterns may also result in a spread of disease and parasites into new regions or produce an increase in the incidence of disease, which, in turn, would reduce animal productivity and possibly increase animal mortality (UNEP, 1998; Hoffmann, 2010).

Maintaining and Improving Environmental Adaptability of Livestock through Genetics

Adaptability refers to the potential or actual capacity to survive and maintain productivity in a wide range of environmental conditions (Hoffmann, 2010). Enhancing adaptation of livestock to climate change and variability needs to involve multiple aspects of animal production such as housing, reproduction, nutrition, health and genetics (Hoffmann, 2010; Boettcher et al., 2014).

Intense artificial selection focused predominantly on production traits has led to degradation of genetic diversity (FAO, 2007; Groeneveld et al., 2010). In addition, the use of a few superior males to inseminate a large number of females has led to a decrease in effective population size in various livestock species and breeds (Kijas et al., 2012). It is essential that breeding programs make efforts to avoid degrading genetic variability since it is directly linked to the potential to cope with adverse conditions such as those caused by climate variability and change (FAO, 2007; Hoffmann, 2010). In addition to maintaining genetic diversity, breeding schemes need to consider improving adaptability traits (e.g., resilience to higher temperatures, lower quality diets and disease pressures) alongside production traits (Åby & Meuwissen, 2010).

Identifying an optimal breeding strategy is an important step towards achieving genetic improvement in adaptation (Boettcher et al., 2014). Available strategies include pure breeding, cross breeding, introgression and breed substitution (Hayes et al., 2013; Boettcher et al., 2014). One of the factors affecting the choice of strategy is the time required for a certain level of genetic change. Cross breeding and breed substitution result in the fastest change (Boettcher et al., 2014). In addition, the use of genomic information (e.g., genomic breeding value) in any of the strategies mentioned above

expedites genetic change (Åby and Meuwissen, 2010). Genome editing (Tan et al., 2013; Proudfoot et al., 2015) could represent an even faster and more direct way of enhancing adaptive capacity, but much work remains in terms of understanding the genomics of adaptation, perfecting gene editing techniques, ensuring food safety and improving public understanding (Bawa and Anilakumar, 2013).

One of the main challenges faced by efforts to improve adaptability in livestock is the negative correlation between certain adaptability traits and production traits. An example of such a relationship is between thermo-tolerance and milk yield in dairy cattle (Hoffmann, 2010). Another related issue when it comes to including adaptability and genetic diversity in breeding programs is the lack of immediate economic benefits for doing so. It has been suggested (Boettcher et al., 2010; Drucker, 2010) that breeders who make such efforts should be given compensation proportional to the expected long-term benefits by a public body. To this end, studies that estimate the overall long term benefits of such programs and examine ways of incentivizing appropriate breeding programs are important (Drucker et al., 2001; Drucker, 2010; Naskar et al., 2012).

Genomics of Adaptive Genetic Variation

Variability in the genetic mechanisms behind adaptability traits (e.g., heat tolerance) between and within different breeds and species of livestock has been well documented by numerous studies (Hayes et al., 2009; Hill and Zhang, 2009; Naskar et al., 2012; Hoffmann, 2013). Further characterization of such variation should provide valuable input to efforts aimed at devising ways to incorporate adaptability traits into genomic breeding programs (Hayes et al., 2013; Boettcher et al., 2014). Genomics is playing (Frichot et al., 2013; Messer and Petrov, 2013; Kim and Rothschild, 2014; Lv et

al., 2014; Kim et al., 2015) and will continue to play a role in the identification of genetic features affecting adaptability traits (Boettcher et al., 2014).

A major challenge when it comes to identifying loci showing signs of adaptive variation is to differentiation them from non-adaptive or selectively neutral loci (Holderegger et al., 2006). According to the theories of neutral or nearly neutral molecular evolution, most of the genetic variation on a molecular level is largely driven by genetic drift and does not display adaptive variation (Nei et al., 2010).

Various methods have been employed to identify parts of the genome affecting environmental adaptation mainly through searching for loci showing signs of natural selection whilst accounting for neutral background genetic variation (Joost et al., 2007; Stapley et al., 2010; Franks and Hoffmann, 2011; Joost et al., 2013; E Frichot et al., 2015). Such methods can generally be divided into two groups. One group consists of population genomics methods (also called outlier tests; Luikart et al., 2003) which identify genomic regions showing high level of differentiation among populations from different environments as compared to a neutral model (Rellstab et al., 2015). However, other than assuming differences in selective pressure between the populations being compared, outlier detection methods do not make a direct link to a specific environmental element as being the underlying cause of selective pressure (Rellstab et al., 2015). A second group of methods, commonly termed environmental association (EA) analysis or landscape genetics analysis, associate allele frequencies directly to an environmental variable (Manel and Holderegger, 2013). Both groups of methods, separately or in combination, have been successfully used in numerous studies to identify loci and candidate genes associated with environmental adaptability (Ramey et

al., 2013; Kijas and Naumova, 2014; Kim and Rothschild, 2014; Lv et al., 2014; Kim et al., 2015).

Materials and Methods

Animals and Genotyping

The study included 184 sheep sampled from 4 of the 8 climatic regions of the US described by the National Oceanic and Atmospheric Administration (NOAA; NOAA, 2013). The four regions were selected to represent a wide range of climatic conditions present in the contiguous US and included Northwestern, Midwestern, Southeastern and Southern parts of the country (Figure 2-1). Northwestern US has one of the highest precipitation rates in the country and is cold for most of the year (Littell, J.S. et al., 2009). Midwestern US is generally characterized by a highly variable climate with cold winters and hot summers (Andresen et al., 2012). Southeastern part of the US is mostly known for high temperature and humidity for a significant portion of the year (Ingram et al., 2013). The climate of Southern US can vary widely through the year, with cold winters and hot and humid summers, although some areas can be quite arid in the summer (LAWSON and STOCKTON, 1981).

The samples from each location included three breeds of sheep, namely: Katahdin, St. Croix and Dorper. Table 2-1 shows the number of each breed per region.

Table 2-1**.** Number of samples per region and breed

Region	Breed		
	Dorper	Katahdin	St. Croix
Midwest	17	19	13
Northwest	11	18	21
Southeast	16	13	15
South	15	14	12

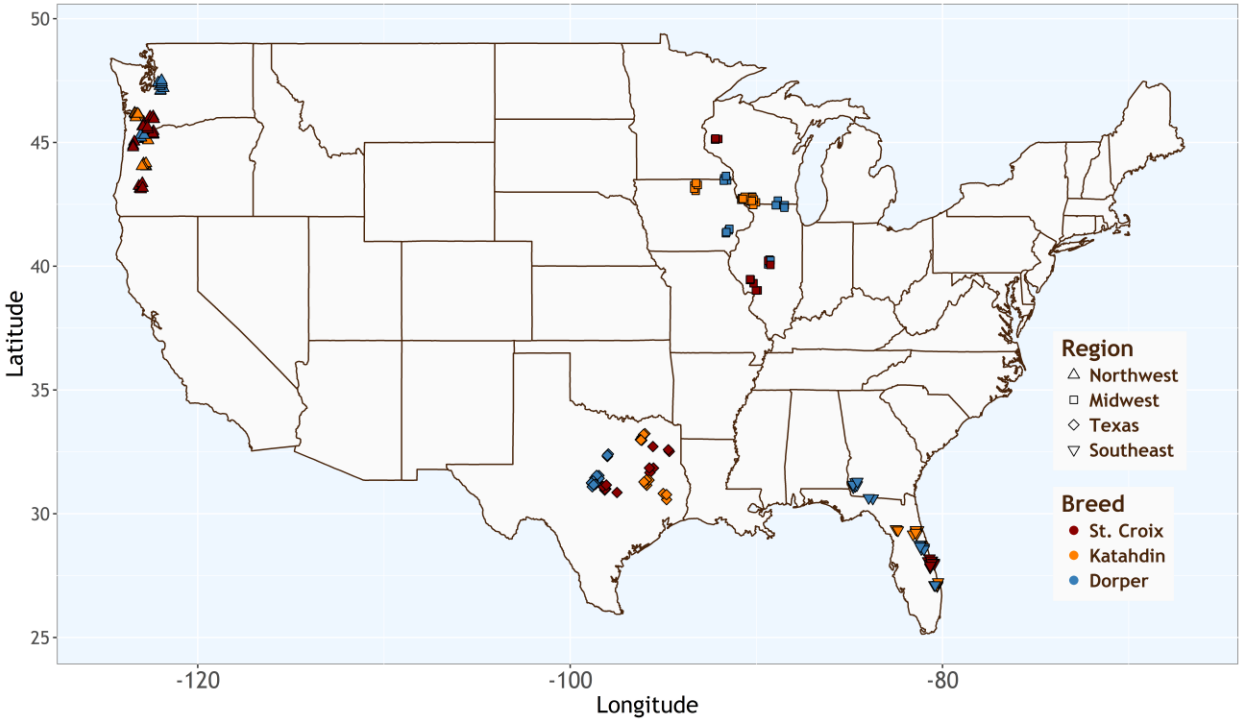


Figure 2-1. The US map showing the sampling locations of sheep in the current study.

An ear punch tissue sample was collected from each sheep using the Allflex Tissue Sampling Applicator ® (Allflex, 2017) which simultaneously transfers the sample into a sealed container (Tissue Sampling Unit ®). The QIAamp® DNA Mini kit (QIAGEN, 2016) was used to extract DNA from tissue samples. The extraction consisted of three main steps. The first step involved lysis of the cells in samples by mixing ~25 mg of tissue with 20 µL of proteinase K in a 2ml Eppendorf tube and incubating at 56 ºC for 8 hours. In the second step, the lysate was transferred to a QIAamp ® mini spin column, which has a silica-based filter membrane that captures DNA molecules. The spin column was then centrifuged, during which DNA molecules were adsorbed by the silica membrane while other components of the lysate passed through. Remaining impurities were removed in two subsequent washing steps. In the final step, the DNA bound to the silica-based membranes was eluted with a buffer solution (10 mM Tris·Cl & 0.5 mM EDTA). The DNA samples were then genotyped using the OvineSNP50 BeadChip

(Illumna, 2015) which is based on sheep genome assembly version 3.1 (Archibald et al., 2010).

Both per-marker and per-animal QC filters were applied to genotype data to avoid potential bias (Anderson et al., 2011). Samples with genotype completion rate of less than 0.8 were excluded from further analysis. In addition, samples having pairwise IBS of >0.98 were considered duplicates and removed from further analysis (Turner et al., 2011). Only autosomal SNP were used in the current study. Monomorphic SNP (i.e., not having an alternative allele in at least one animal) and those with call rate of less than 0.95 were also removed from analysis (Turner et al., 2011). Furthermore, LD pruning was performed by defining a window size of 1000 kbp and an LD threshold of 0.5 (Anderson et al., 2011). All QC measures were applied using PLINK1.9 (Chang et al., 2015a).

Environmental Data

Climatic conditions of the sampling locations were characterized by summarizing daily measurements of five environmental variables for the time between 1994-01-01 and 2014-01-01. The variables were altitude, precipitation (PRCP), daily maximum temperature (TMAX), daily minimum temperature (TMIN) and daily temperature humidity Index (THI). Daily THI was calculated from average daily temperature and daily relative humidity (RH) using the formula in Dikmen *et al* (2013). All environmental data was obtained from the Global Surface Summary of the Day (GSOD) database by the National Climatic Data Center (NCDC) of the US (NOAA, 2017). The database was accessed using the R package GSODR (Adam Sparks et al., 2017a). In addition to downloading weather data from the NCDC GSOD FTP server

(ftp://ftp.ncdc.noaa.gov/pub/data/gsod/), the GSODR package also does data cleaning and reformatting.

Retrieving environmental data

Environmental data was obtained from weather stations selected based on three criteria: presence of data for the time between 1994-01-01 and 2014-01-01, proximity to the sampling locations, and proportion of missing data. A list of all available weather stations in the US with data in the GSOD database was downloaded from the NCDC FTP server using the 'get_stations_list' function in the GSODR package. Stations that were not in or near the contiguous US and stations that did not have data for the time between 1994-01-01 and 2014-01-01 were filtered out. Pairwise distance (in kilometers) between sampling locations coordinates and station coordinates was calculated using the Haversine formula (Sinnott, 1984) implemented the R package 'geosphere' (Hijmans, 2016).

For a given sampling location, environmental data recorded in the closest shortlisted station was retrieved from the NCDC GSOD FTP server (ftp://ftp.ncdc.noaa.gov/pub/data/gsod/) using the function 'get_GSOD' in the GSODR package. If the proportion of missing values in the retrieved data was less than 10%, it was used for subsequent analysis. However, if the proportion of missing values was greater than 10%, data was retrieved again from the next closest station. This process was repeated until data with less than 10% missingness was found or the fifth closest station was reached. If none of the five closest stations had data with less than 10% missingness, data with the least missingness from these five station was used in subsequent analysis. An algorithm (Appendix) was constructed using the R programming language (R Core Team, 2016) to perform the steps mentioned above.

Summarizing environmental data

For all environmental variables except altitude, daily data were converted to monthly averages. Therefore, 961 variables (i.e., 20 years of monthly averages of four variables in addition to altitude) were used in further analysis to characterize the sampling locations. Since most of the weather variables were likely to be highly correlated for a given location, the information they provide could be redundant if all of them were used in a model at the same time (Lv et al., 2014). Therefore, PCA was performed on the standardized environmental data (i.e., with each variable mean-centered and divided by its standard deviation) to extract relevant and non-redundant information. The first PC was then used to characterize the sampling locations in terms of their climatic condition. The absolutes value of the PCA loadings for each environmental variable were used to measure its influence on the first PC and, consequently, on the frequency of SNP associated with the first PC. Principal component analysis was performed using the function ‘prcomp’ in the R package ‘stats’ (R Core Team, 2013).

Genome-wide Environmental Association Analysis

Latent factor mixed model

In order to identify loci associated directly with environmental variables, an EA analysis (Lowry, 2010; Rellstab et al., 2015) was performed using Latent Factor Mixed Model (LFMM) implemented in the ‘lfmm’ function from the R package LEA (Eric Frichot et al., 2015). The model tests for associations between allele frequencies and environmental variables while accounting for background neutral genetic variation by including latent factors derived from genome-wide data as a random effect covariate (Frichot et al., 2013). The LFMM model was described as

$$G_{il} = \mu_l + \beta_l^T X_i + U_i^T V_l + \epsilon_{il}, \quad (2\text{-}1)$$

where G_{il} is a vector of genotypes in locus l for i animals, μ_l is the locus specific effect, β_l is a vector of regression coefficient for environmental variables, X_i is a vector of environmental variable values, U_i is a vector of regression coefficient for each latent factor, V_l is a vector of latent factors, and ϵ_{il} represents independent and normally distributed residuals (Frichot et al., 2013).

The LFMM algorithm simultaneously estimates $\mu, \beta, U \; and \; V$ using Markov Chain Monte Carlo (MCMC), with the number of latent factors (K) being supplied *a priori* (Frichot et al., 2013). The model was run with 5,000 burn-in and 10,000 iterations sweeps. Each run was replicated 5 times, and the median of the z-scores from the five runs was used to make inferences. For each locus, a z-score was computed by dividing the centered value of β_l (i.e., $\beta_l - \hat{\beta}_l$) to the standard deviation of β_l (Frichot et al., 2013). The squared z-scores were used as a test statistic, and p-values were obtained based on a chi-squared distribution with a degree of freedom of one.

Loci with significant association to the environmental variable were identified as having a p-value greater that a threshold established based on the Benjamini–Hochberg (BH) algorithm (Benjamini and Hochberg, 1995a; Francois et al., 2016). The BH algorithm was used with the goal of keeping the rate of false discovery (FDR) below 10%. To obtain BH thresholds, the loci were ordered based on their p-value from smallest to largest, and an index number was assigned such that a locus with the smallest p-value was given an index number of one and the one with the largest p-value was given an index number equal to the total number of loci used in LFMM. For a given locus, its BH threshold was obtained by multiplying 0.1 (i.e., the 10% FRD upper limit) to

the ratio of its index to the total number of loci used in LFMM (Benjamini and Hochberg, 1995b).

The functional consequence of SNPs associated with environmental variables was further explored using the Variant Effect Predictor (VEP) tool (McLaren et al., 2016) available on the Ensemble website (http://useast.ensembl.org/index.html). Based on annotation information for the Ovine v3.1 genome assembly, a 20kbp window (10,000bp upstream/downstream) around the genomic position of each significant SNP was searched for effects on transcripts, proteins, and regulatory regions. Genes affected by environment associated SNPs based on functional annotation were considered candidate genes.

Finding the optimal number of latent factors

Similar to genome wide association studies, the presence population structure can lead to an inflation of type I errors in EA analyses (De Mita et al., 2013). In LFMM, background genetic variation or population structure is accounted for by including latent factors (V_l) inferred from genome-wide data as covariates in the model as random effect. The efficiency at which LFMM accounts for population structure is dependent on the number of K specified (Frichot et al., 2013). A value for K that is too low will inflate false positive results (Type I error) whereas a K number that is too high will inflate the amount of false negative result (Type II error) (Frichot et al., 2013). Therefore, choosing the optimal value for K is a crucial step in LFMM analysis.

Two main steps were taken to identify the optimal value for K. In the first step, an unsupervised clustering was performed on the genotyped data using sparse non negative matrix factorization and least square optimization (E. Frichot et al., 2014) implemented in the function ‘sNMF’, which is part of the LEA R package (Frichot and

Franc, 2015). This clustering algorithm is very similar to the one used in ADMIXTURE, and it estimates cluster membership coefficients and cluster allele frequencies which correspond to the *Q* and *F* matrices of ADMIXTURE. The sNMF program was run separately for eight different values for K (where $K \in \{3, 4, \ldots, 10\}$). For each sNMF run, a cross validation procedure was performed to evaluate how well the K value fits the genotype data (E. Frichot et al., 2014; Shringarpure et al., 2016). For the cross validation step, 5% of the genotype data were randomly selected and tagged as missing. Genotypes for the tagged SNP were then predicted using '*Q*' and '*F*' estimates obtained for the rest of the (untagged) genotype data. The agreement between actual and predicted genotypes for the tagged SNP was measured in terms of a cross-entropy criterion as described by Frichot et al (E. Frichot et al., 2014). The sNMF run for each K was replicated 10 times, and the median of the cross-entropy criterion from these replicates was used to compare between different values of K. A low cross-entropy criterion was considered to indicate a K value that would allow sufficient control of population structure if used in LFMM (Francois et al., 2016).

The second approach that was used in the identification of the optimal number of K was based on genomic inflation factor (GIF) values calculated from results of LFMM runs using different putative values for K (Francois et al., 2016). The GIF for an output of a given LFMM run was obtained by dividing the median of the squared z-scores to the median of a chi-square distribution with a degree of freedom of one. A GIF value of less than or close to one indicated that population structure has been properly accounted for whereas values much higher than one indicate higher number of false positive results due to of population structure (E Frichot et al., 2015).

To appreciate overall population structure and help in the identification of the ideal number K, PCA was performed on genome wide data using PLINK. In addition, pairwise kinship was calculated using the KING-robust method (Manichaikul et al., 2010) to characterize close familial relationships among sheep from nearby sampling locations.

Gene Ontology Term Enrichment

To test if the candidate genes identified through EA analysis show a pattern of functional relation, a Gene Ontology (GO) term enrichment analysis (Morota et al., 2015) was carried out using the PANTHER platform (Mi et al., 2017). Gene ontology relates to the hierarchical classification of genes based on their functions into distinct categories (i.e., terms) organized under three major domains, namely: biological process, molecular functions or cellular components (Harris et al., 2008). Such classification is provided by the PANTHER (protein annotation through evolutionary relationship) Classification System (Mi et al., 2013). Gene Ontology term enrichment analysis is a statistical test to determine if there is an over-representation of a set of candidate genes in one or more GO term categories, as compared to what would be expected for a randomly selected set of genes (Morota et al., 2015). For a given GO term category, the probability of a randomly selected gene to be assigned to it is given by the ratio of the total number of genes in that category to the total number of genes in a reference genome (Mi et al., 2017). Since the sheep reference genome was not available in the PANTHER database, the bovine (*Bos taurus*) genome was used as a reference to obtain the expected probability for a randomly selected gene to be placed in each GO terms category. To enable mapping of the sheep candidate genes on the bovine reference genome, bovine genes that are orthologous to the candidate genes

were used for the enrichment analysis. Orthologous genes were obtained using the BioMart tool (Smedley et al., 2015) available on the Ensembl website (http://useast.ensembl.org/index.html).

After uploading a list of candidate gene symbols to the PANTHER web interface (http://www.pantherdb.org/), GO term enrichment was carried out by the PANTHER tool as follows. The first step was to categorize the list of candidate genes into different GO terms. For each category, a binomial test was performed to see if there was an over-representation of the candidate as compared to what would be expected if the genes were to be selected randomly from a reference genome. A chi-squared test p-value of 0.05 was used as a threshold to identify GO terms with an overrepresentation of candidate genes.

Visualization of Results

Proper visualization is an important part of communicating study results. Freely available packages in the R programming environment were used to help illustrate the results obtained in the current study. These packages were: 'ggplot2' (Wickham, 2009) 'base' (R Core Team, 2016), and 'qqman' (Turner, 2017).

Results and Discussion

Genotype Quality Control

After removing one sample for failing to meet the genotype completion rate criteria (i.e., < 0.8) and a pair of samples for being duplicates (i.e., IBS of > 0.98), 181 samples were kept and used in subsequent analysis. From an initial set of 54,241 SNP, a subset of 43,118 SNP was kept after excluding 725 monomorphic SNP, 1,599 SNP that failed to meet the minimum call rate threshold (i.e., < 0.95) and 6,971 SNP due to LD pruning. Non-autosomal SNP (n = 1,828) were also not included in the current study.

Therefore, a total of 43,118 SNP and 181 sheep passed QC, and they were used in subsequent analysis.

Environmental Data Retrieval and Summary

From 28,327 GSOD stations, 3115 were identified as being in or near the contiguous US (i.e., within latitudes 25 and 50 and longitudes -125 and -60) and having data for the time between 1994-01-01 and 2014-01-01. Pairwise distance between the coordinates of all shortlisted stations and all sampling locations was constructed using the Haversine formula implemented in the R package GSODR r (Adam Sparks et al., 2017b). For each sampling location, the closest station with minimal missing data was identified using an algorithm described in the Materials and Methods section. Twenty-two stations across the US were identified based on the criteria mentioned above (Figure 2-2).

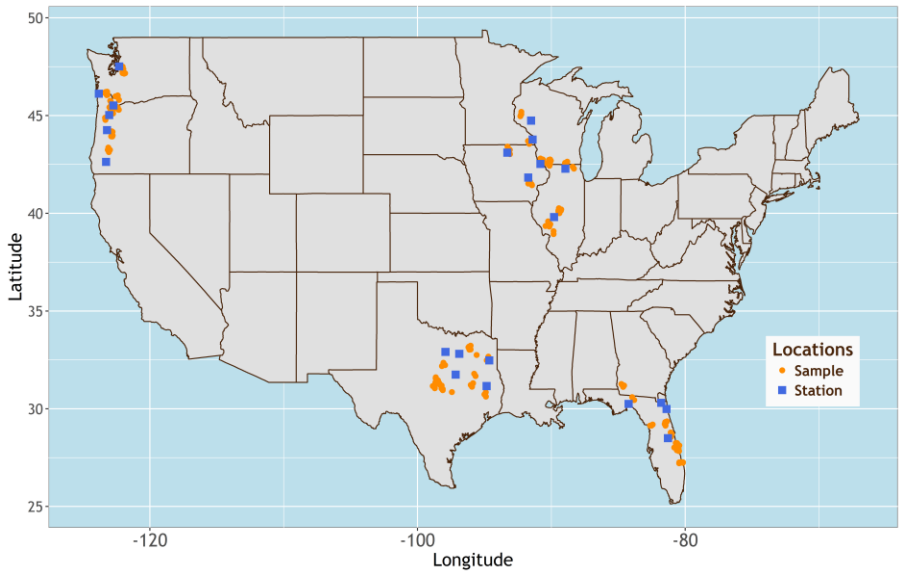


Figure 2-2. A map showing sampling locations and locations of stations from which data was retrieved.

There was strong positive correlation among MAX, MIN and THI. Precipitation had a moderate positive correlation with MAX, MIN and THI. In contrast, altitude had a

moderately negative relationship between some of the environmental variables and a moderate to weak level of correlation between the rest of the variables. This can be seen in Figure 2-3, which shows a heat map based on the correlation matrix between the 5 environmental variables. To identify the main axis of variation and extract non-redundant information, PCA was applied to 961 environmental variables (i.e., 20 years of monthly averages of 4 variables in addition to altitude).

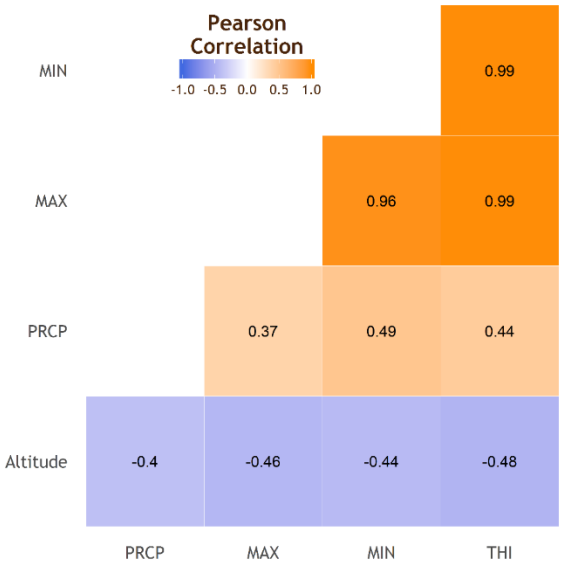


Figure 2-3. A heat map showing the relationship between the 5 environmental variable considered. For all variables except altitude, the average of 20 years of daily values was used to compute the correlation matrix used to make the heat map.

From PCA on 20 years of monthly average values for the 5 environmental variables, the first and second PC explained 66.6% and 15.6% of the total variation, respectively. On a scatterplot of PC1 and 2 (Figure 2-4), the data points for the sampling locations clustered in manner consistent with the 4 regions of the US from which samples were obtained.

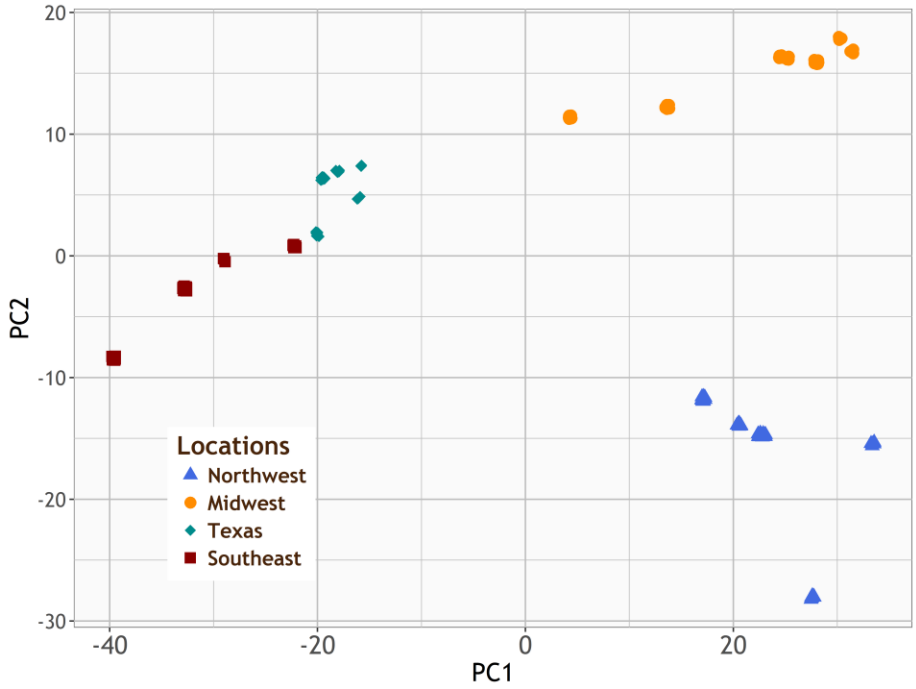


Figure 2-4. A scatterplot showing the position of the sampling locations respective to the first 2 top PC.

To evaluate the influence of individual environmental variables on the top PC, the squared sum of their PCA loadings were compared. In other words, for a given variable, its PC loading coefficients for 20 years of monthly averages were squared and added. These values were then compared among the 5 variables. There was only one loading coefficient per PC for altitude, which had the lowest influence on both PC (0.0003 and 0.0004 for PC1 and 2, respectively). Temperature humidity index had a slightly higher influence on PC1 (0.3231) as compared to MAX (0.3074) and MIN (0.2996) whereas PRCP had the second lowest influence (0.0694). Interestingly, PRCP had the highest influence on PC2 (0.3814) whereas MIN (0.2460), THI (0.2038) and MAX (0.1682) were 2nd, 3rd and 4th respectively. Since PC1 explained the majority of the variation, it was used in the LFMM model to represent the fixed effect of the environment (Lv et al., 2014).

Genome-wide Environmental Association Analysis

Finding the optimal number of latent factors

The first criteria used to identify the ideal number of K for use in LFMM was the cross-entropy criterion calculated by sNMF (E. Frichot et al., 2014). The sNMF software was run for values of K ranging from 3 to 10. The run was replicated 5 times for each value of K. The median of the cross-entropy criterion from the 5 replicates was then used to compare the different K values. Figure-4 shows the result from this procedure. It can be seen that a K value of 6 had the lowest cross-entropy criterion. This result indicated that the ideal number of K that would allow proper control of population structure in LFMM is 6 (Eric Frichot et al., 2015).

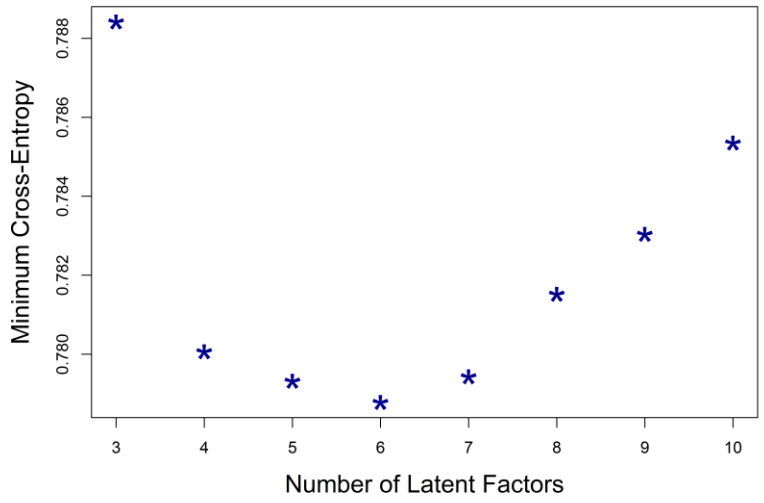


Figure 2-5. A scatterplot showing the result from the cross-validation procedure of sNMF run on 8 different values of K.

To further evaluate whether a K value of 6 was sufficient to control for population structure, GIF was calculated after running LFMM on whole genome data (post QC) using 6 latent factors. Genomic inflation factor was calculated by dividing the median of the squared z-scores for all SNP to the median of a chi-squared distribution with a

degree of freedom of 1. From this LFMM run, a GIF value of 1.85 was obtained. This meant that the expected z-score for neutral loci was 1.85 (corresponding to a p-value of 0.064) , which indicated that a significant amount of population structure was still confounding LFMM results (Francois et al., 2016). Therefore, LFMM was run again using 8 latent factors, with the other parameters remaining the same, and a GIF value of 1.21 (corresponding to a p-value of 0.23 for the expected test statistic for neutral loci) obtained. This indicated that eight latent factors allowed sufficient control population structure (Francois et al., 2016). Therefore, LFMM results from this run (i.e., using eight latent factors) were used for inference on association between loci and environmental variables.

The fact that eight latent factors were required to achieve a reasonable control of population structure would have been surprising if the only source of population structure was breed difference. If that was the case, the number of K with the lowest cross-entropy criterion should have been three (Pritchard et al., 2000; Patterson et al., 2006). However, genomic data of the study population was expected to reveal multiple levels of population structure coming from diverse sources.

In addition to there being sheep from 3 different breeds, there was also most likely to be some level of within-breed differentiation between sheep sampled from different regions of the US. Population subdivision (e.g., due to spatial separation) resulting in a reduce ability to interbreed could lead to the creation of genetic structures specific to subpopulations, even in the absence of forces other than genetic drift (Wright, 1951; Roughgarden, 1979). The degree of differentiation between the subpopulations would of course depend on numerous factors including the extent of

separation, rate of migration between subpopulations, number of generations elapsed since separation, amount of genetic variation, intensity of selective pressure (if any), size of the total population and size of the subpopulations (Roughgarden, 1979; Zhivotovsky, 2015).

Another source of population structure was familial relationship between sheep sampled from nearby locations or the same farm. This can be appreciated from Figure 2-6, which shows pairwise kinship and divergence measurements estimated from genetic data using the KING-robust method (Manichaikul et al., 2010). Two things can be observed in this figure. One is that there are several close familial relationships. The other is that there is a large amount of divergence (Matthew P Conomos et al., 2016) among samples, indicating the presence of strong population structure.

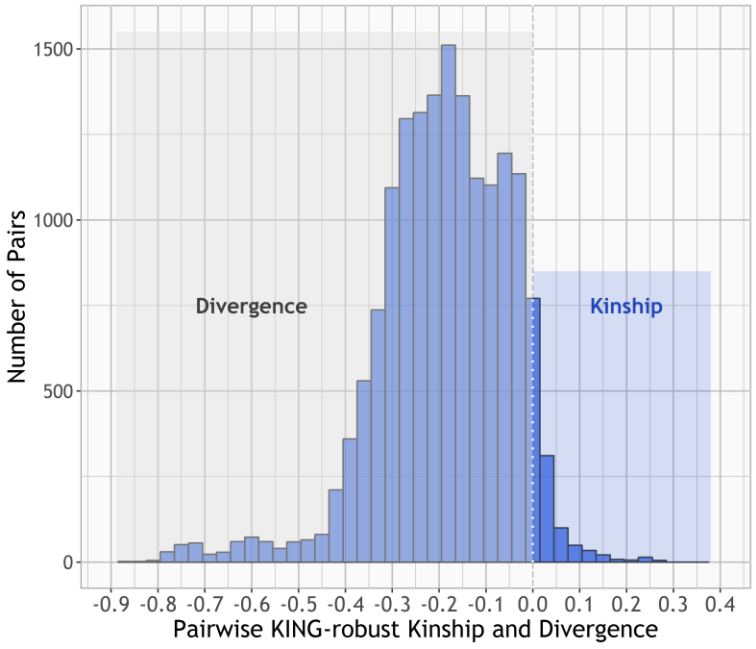


Figure 2-6. Histogram showing pairwise KING-robust kinship and divergence estimates for all sheep. There were a number of close familial relationships as can be seen from the kinship estimates. The large number of divergence estimates indicates the presence of very strong structuring of genetic variation.

The study population had a complex population structure resulting from the interaction between at least three factors, namely: breed difference, familial relationships and spatial separation. The combined effect of these factors can be appreciated in Figure 2-7, which shows a plot between PC1 and 2 from PCA applied to whole genome data (post QC). Even though the sheep formed three clusters based on their breeds, it was also evident that certain samples tended to converge towards the center. This tendency was more prominent among sheep from Southeastern US and, to a lesser extent, those from the Midwest. In contrast, sheep from the Northwest tended to be positioned away from the center. The patterns seen here are likely due to the superimposition multiple factors including spatial separation, breed ancestry, and recent ancestry or familial relationships (Roughgarden, 1979; Patterson et al., 2006).

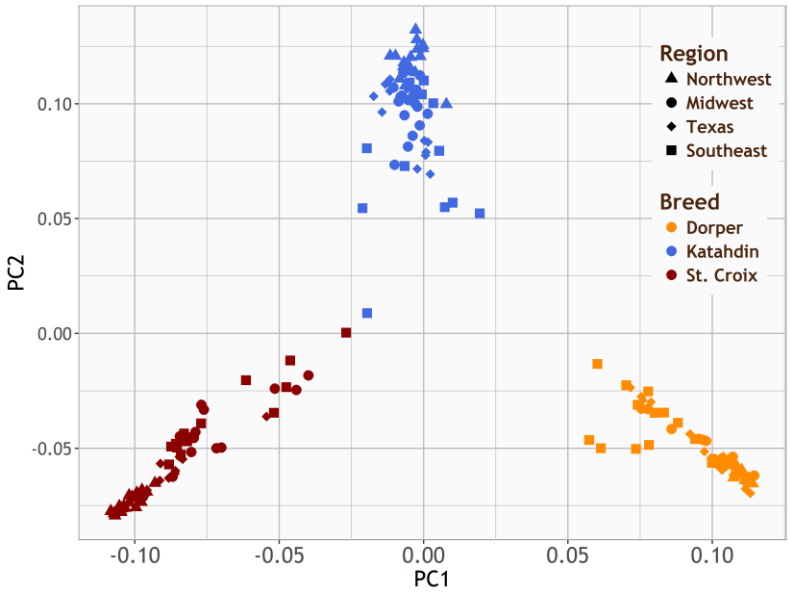


Figure 2-7. A PC1 versus PC2 scatterplot from PCA applied to genotype data showing overall population structure among all sheep in the study (n=181).

Environmental association analysis

As described earlier, LFMM was applied to genomic data using PC1 from PCA on environmental data as a fixed environmental effect, eight latent factors as random effects. The model was run using 5,000 burn-in and 10,000 iterations sweeps. After obtaining z-scores for each locus, p-values were obtained using a chi-squared distribution with a degree of freedom of one. To keep FDR under 10%, significance thresholds for each SNP were obtained using the BH method (Benjamini and Hochberg, 1995a).

In EA analyses, most loci are expected to be selectively neutral, and only a small minority are expected to show evidence of adaptive variation (Francois et al., 2016). This is in line with the neutral theory of molecular evolution which suggests that, on a molecular level, most variations are selectively neutral and vary according to the rules of random genetic drift (Nei et al., 2010). Therefore, in a well calibrated EA analysis (i.e., properly accounting for confounding from population structure or background neutral variation), p-values for most loci should follow a uniform distribution between 0 and 1, whereas a small proportion of the markers typically show a frequency peak near 0 (Francois et al., 2016). Figure 2-8 shows the distribution of p-values from LFMM across all loci, which is consistent with the description of a properly calibrated EA analysis by Francois et al (Francois et al., 2016).

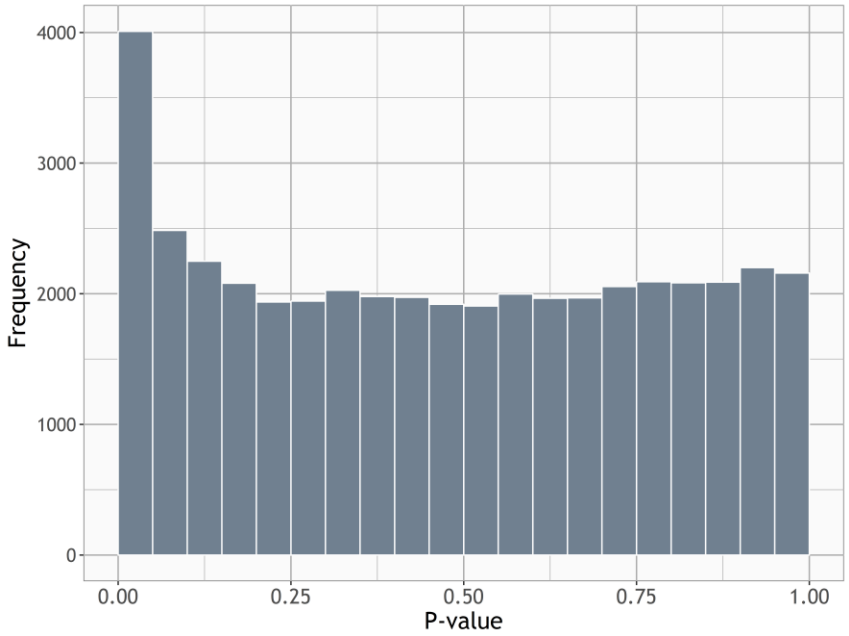


Figure 2-8. Histogram showing distribution of p-values from LFMM. The uniform distribution of neural (high p-value) loci is indicative of a well-calibrated genomic scan for adaptive loci (Francois et al., 2016)

Loci with significant association to environmental variation were identified as having a p-value greater than a BH threshold (Benjamini and Hochberg, 1995a; Francois et al., 2016). There were 389 SNP that had a p-value above BH cut off. Significant SNP were distributed throughout the genome, although most of the significant SNP were in chromosomes one (n = 48), two (n = 34) and three (n = 40). Chromosome 24 (n=1) had the lowest number of significant SNP. The overall distribution of p-values across the genome is shown in Figure 2-9 which was produced using the 'qqman' R package (Turner, 2017) . The genome wide distribution of SNP with low p-values can be appreciated in this Manhattan plot. This is consistent with the expectation that the genetics underlying environmental adaptation is highly complex, involving many genes (Franks and Hoffmann, 2012; Hayes et al., 2013; R.I.C. Frichot et al., 2014).

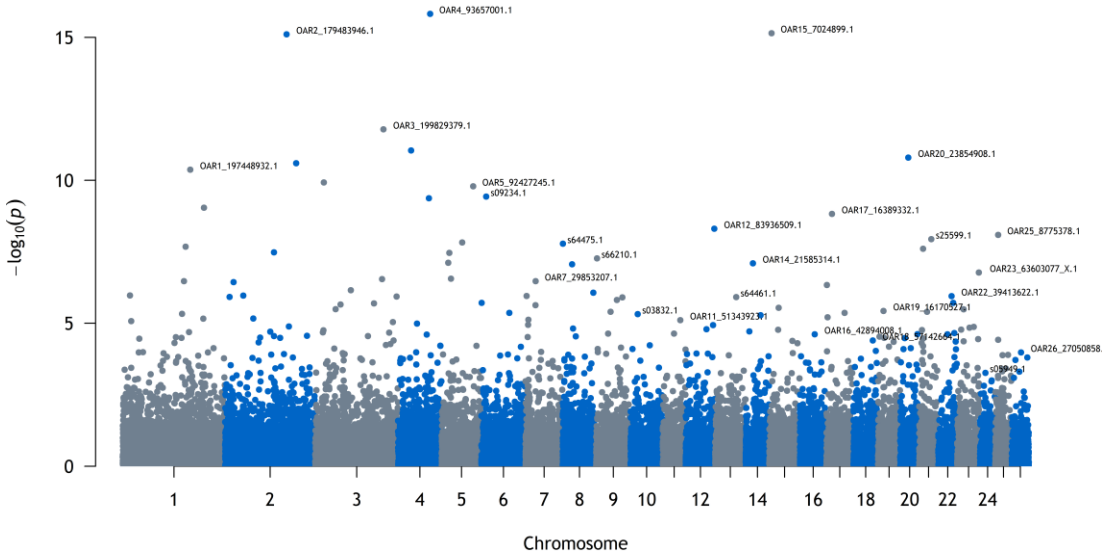


Figure 2-9. Manhattan plot showing negative log of p-values from LFMM for loci across the genome. For each chromosome, one SNP with the highest negative log of p-value was labeled with its variant ID.

Functional annotation of environment-associated SNP was performed using the VEP tool (McLaren et al., 2016) made available on the Ensembl website (http://useast.ensembl.org/index.html). While the majority (n= 203) of environment-associated SNP were intergenic variants, the rest (n = 186) overlapped with 184 genes. Of these, the most consequential were three missense variants (1:21200152 G/T; 17:59968700 T/C and 21:44780762 A/G) that result in amino acid sequence changes. In addition, there were 126 intronic, 28 upstream and 26 downstream variants. Some SNP were assigned more than functional annotation terms.

Gene Ontology Enrichment

To evaluate whether there is any functional association between candidate genes, GO term enrichment analysis was carried out using the PANTHER web interface tool (Mi et al., 2017). Since the bovine genome was used as a reference, the bovine orthologues all sheep candidate genes were used for the enrichment analysis. Out of

the 184 genes that overlapped with the environment-associated SNP, a bovine orthologue was found for 173 genes.

Candidate genes were evaluated for all three major GO categories for functional enrichment using a significance threshold of 0.05 (Lv et al., 2014). Under biological processes, there was overrepresentation of candidate genes in five subcategories, namely: protein lipidation, female gamete generation, cellular defense response, nuclear transport, and heart development. Within the domain of molecular functions, five main terms showed overrepresentation of candidate genes. These were RNA-directed DNA polymerase activity, voltage gated ion channel activity, RNA methyl transferase activity, enzyme regulation activity and protein binding. There was no overrepresentation of GO terms under the cellular components domain.

The enrichment results revealed a wide range of physiological processes affected by the candidate genes including, immunity, reproduction and cardiovascular functions. Considering the broad range of traits underlying the ability of an organism to survive and thrive in a given environment (Franks and Hoffmann, 2012), it is expected for a set of genes associated with an environmental gradient to have a diverse functional enrichment (Manel et al., 2010; Somero, 2010; Lv et al., 2014). However, such results be interpreted with caution (Joost et al., 2013), and validating results using multiple approaches is warranted.

Conclusion

The climatic conditions of the sampling locations of the study animals were characterized by summarizing 20 years of climate data for five environmental variables. From PCA applied to environmental data, PC1 explained most of the variation (66.6%), and it was subsequently used as a proxy for environmental variation in EA analysis.

Using LFMM, it was possible to identify numerous loci associated with environmental variation after properly accounting for population structure or background neutral genetic variation. The presence of multiple hierarchies of population structure in the study population offered both a challenge and an opportunity. On one hand, it warranted the careful application of different measures (e.g. the use of GIF and cross-entropy criterion) to account for background genetic variation when identifying loci associated directly with environmental variables. On the other hand, it offered a chance to study environmental effect on genetic composition across breeds and ecotypes. This prospect was augmented by the well-balanced sampling scheme that enabled fair representation of sheep from different breed/ecotypes and a wide range of climatic regions of the US (Table 1).

From LFMM analysis, 389 environment-associated SNP were identified based on BH threshold intended to keep FDR at or below 10%. The genome-wide distribution of significant SNP is inline the widely-accepted notion that the genetics behind environmental adaptation is highly complex, involving numerous loci across the genome. Of the significant SNP, 186 overlapped with 184 candidate genes. Functional annotation of significant SNP overlapping with genes led to the identification of missense (n = 3) intronic (126), upstream (28) and downstream (26) variants.

Gene ontology enrichment test identified overrepresentation of candidate genes in GO term categories related to a wide range of biological processes and molecular functions.

The results obtained in the current study are encouraging and open the door for more detailed analyses. Further characterization of adaptive variation using methods

such as outlier detection test, integrated haplotype score and window based screening for selective sweeps will complement the results obtained here.

APPENDIX
R CODES

The following is an R code for cross-validation of breed composition prediction accuracy.

```
# Description of function arguments:- 1. dgdata: genotype data were the
# first column has phenotype values, and the rest of the columns have SNP
# data (coded 0,1,2) 2. k: the number of 'folds' for a k-fold
# crossvalidation, (i.e. the number of groups the data will be divided into)
# 3. coln: the number of SNP columns to use. This is useful if this function
# is put in a loop to perform the crossvalidation for different number of
# SNP. If for example 'coln' is 10, only the first 10 SNP are used. In my
# case SNP are ordered based on their their contribution to population
# structure, so the first 10 SNP are the top 10 SNP with strongest
# contribution to population structure.  4. sub: this is a logical(T/F)
# specifying weather to use a subset of SNP for crossvalidation. If True,
# 'coln' has to be specified and vice versa. The default is False.

# function output: The function outputs 'k' correlation values for a k-fold
# crossvalidation

function(gdata, k = 5, coln, sub = F) {

    # if 'sub' is true, select 'coln' SNP columns from gdata
    if (sub) {
        gdata <- gdata[, c(1:(coln + 1))]
    }

    # randomize the order of rows
    gdata <- gdata[sample(nrow(gdata)), ]

    # add a grouping variable with k groups to gdata
    let <- letters[1:k]
    group <- rep(let, each = floor(nrow(gdata)/k), length.out = nrow(gdata))
    gdata <- cbind(group, gdata)

    # do k-fold cross validation and save 'k' correlation/accuracy values in
    # 'correlations'
    correlations <- sapply(let, function(x) {
        train_d <- gdata[!(gdata$group == x), -1]
        pred_d <- gdata[gdata$group == x, -1]
        m <- lm(Angus ~ ., data = train_d)
        cor(predict(m, newdata = pred_d[, -1]), pred_d[, 1], use = "complete.obs",
            method = "pearson")
    })
    return(correlations)
}
```

The following is a code for retrieving environmental data as described in the Materials and Methods section of the first chapter. Further details about the code can be found in the following GitHub repository: https://github.com/Mesfingo/Download-and-summerize-historical-climate-data

```
variable_data <- function(variable, location, year_range, distance_matrix) {
    if (!suppressMessages(require("GSODR", quietly = T))) {
        install.packages("GSODR")
    }
    variable_dat <- lapply(1:length(location), function(i) {
        # cache missingness data for stations
        missingness_cache <- vector(length = 5)
        # sort stations by distance
        closest_station <- sort(distance_matrix[location[i], ], decreasing = F)
        condition <- TRUE
        for (j in 1:5) {
            # get name for closest station
            closest_station_name <- names(closest_station[j])
            # download closest station data
            station_data <- get_GSOD(years = year_range, station = closest_station_name,
                country = "US")
            station_data <- station_data[station_data[["STNID"]] %in% closest_station_name,
                ]
            # get data quality measures how many rows (days) does the data have?
            is_data_small <- dim(station_data)[1]
            is_variable_there <- variable %in% names(station_data)  # variable of interest there?
            is_missing <- round(sum(is.na(station_data[[variable]]))/length(station_data[[variable]]),
                6)  # what percent of data is missing?
            missingness_cache[j] <- is_missing  # keep track of missingness for a given cycle (1:5)
            names(missingness_cache)[j] <- closest_station_name  # name cache value by station
            # are all conditions NOT met?
            condition <- ((is_data_small < (360 * (length(year_range))) || !is_variable_there ||
                is_missing > 0.1) && j <= 5)
            # if true and 5th station is reached, download data from the one with least
            # missingness of the 5 closest stations
            if (condition == TRUE & j == 5) {
                closest_station_name <- names(which.min(missingness_cache))
                station_data <- get_GSOD(years = year_range, station = closest_station_name,
                  country = "US")
                station_data <- station_data[station_data[["STNID"]] %in% closest_station_name,]
                break
            }
            # if all conditions are not met and 5th station is not reached, move to next
            # closest station
            if (condition == TRUE) {
                next
            } else {
                break
            }
        }
        # combine additional information (e.g., distance from location) to
        # downloaded data
        location <- rep(loc[i], nrow(station_data))
        distance <- distance_matrix[location[i], c(closest_station_name)]
        distance_from_location <- rep(distance, nrow(station_data))
        station_data <- cbind(station_data, location, distance_from_location)
        return(station_data)
    })
    # return a list whose elements contain data for 1 or more location
    return(variable_dat)
}
```

LIST OF REFERENCES

Åby, B. A., and T. Meuwissen. 2010. Selection strategies utilizing genetic resources to adapt livestock to climate change. In: 10th World Congress on Genetics Applied to Livestock Production. p. 2–4.

Adam Sparks, Tomislav Hengl, and Andrew Nelson. 2017a. GSODR: Global Summary Daily Weather Data in R. Available from: https://github.com/ropensci/GSODR

Adam Sparks, Tomislav Hengl, and Andrew Nelson. 2017b. GSODR: Global Summary Daily Weather Data in R.

Ajmone-Marsan, P., J. F. Garcia, J. A. Lenstra, and C. Globaldiv. 2010. On the Origin of Cattle: How Aurochs Became Cattle and Colonized the World. Evol. Anthropol. 19:148–157. doi:10.1002/evan20267.

Akerman, J. 1982. American Brahman: A History of the American Brahman. 1st ed. Amer Brahman Breeders Association.

Alexander, D. H., and K. Lange. 2015. Admixture 1.3 Software Manual. 3–4.

Alexander, D. H., and J. Novembre. 2009. Fast Model-Based Estimation of Ancestry in Unrelated Individuals. 1655–1664. doi:10.1101/gr.094052.109.vidual.

Alexander, D. H., J. Novembre, and K. Lange. 2009. Fast model-based estimation of ancestry in unrelated individuals. 1655–1664. doi:10.1101/gr.094052.109.vidual.

Allflex. 2017. NextGen Tissue Sampling Technology by Allflex - Allflex » Allflex USA. Available from: http://www.allflexusa.com/our-products/cattle/product/allflex-tissue-sampling-technology

American Angus Association. 2016. Angus FAQs. Available from: https://www.angus.org/Pub/FAQs.aspx

Anderson, C. a, F. H. Pettersson, G. M. Clarke, L. R. Cardon, P. Morris, and K. T. Zondervan. 2011. Data quality control in genetic case-control association studies. Nat. Protoc. 5:1564–1573. doi:10.1038/nprot.2010.116.Data.

Andresen, J., S. Hilberg, and K. Kunkel. 2012. Historical Climate and Climate Trends in the Midwestern USA. Available from: http://glisa.msu.edu/docs/NCA/MTIT_Historical.pdf

Archibald, A. L., N. E. Cockett, B. P. Dalrymple, T. Faraut, J. W. Kijas, J. F. Maddox, J. C. McEwan, V. Hutton Oddy, H. W. Raadsma, C. Wade, J. Wang, W. Wang, and X. Xun. 2010. The sheep genome reference sequence: A work in progress. Anim. Genet. 41:449–453. doi:10.1111/j.1365-2052.2010.02100.x.

Backlund, P., A. Janetos, and D. Schimel. 2008. The Effects of Climate Change on Agriculture , Land Resources , Water Resources , and Biodiversity in the United States.

Available from: http://www.climatescience.gov

Barros, V., T. F. Stocker, D. Qin, D. J. Dokken, K. L. Ebi, M. D. Mastrandrea, K. J. Mach, S. K. Allen, and M. Tignor. 2012. Glossary of terms In: Managing the Risks of Extreme Events and Disasters to Advance Climate Change Adaptation. . A Spec. Rep. Work. Groups I II Intergov. Panel Clim. Chang. 555–564. doi:10.1177/1403494813515131. Available from: http://www.ncbi.nlm.nih.gov/pubmed/24819604

Bauck, S. 2016. Where are we going with bisphosphonates? Progress. Cattlem. Available from: http://www.progressivecattle.com/topics/reproduction/7539-where-are-we-going-with-genomics

Bawa, A. S., and K. R. Anilakumar. 2013. Genetically modified foods: Safety, risks and public concerns - A review. J. Food Sci. Technol. 50:1035–1046. doi:10.1007/s13197-012-0899-1.

Beddington, J., M. Asadazzama, M. Clark, A. Fernández, M. Guillou, M. Jahn, L. Erda, T. Mamo, N. Van Bo, C. a Nobre, R. Scholes, R. Sharma, and J. Wakhungu. 2012. Achieving food security in the face of climate change. Available from: www.ccafs.cgiar.org/commission

Benjamini, Y., and Y. Hochberg. 1995a. Controlling the False Discovery Rate : A Practical and Powerful Approach to Multiple Testing Author (s): Yoav Benjamini and Yosef Hochberg Source : Journal of the Royal Statistical Society . Series B (Methodological), Vol . 57 , No . 1 Published by : J. R. Stat. Soc. Ser. B Stat. Methodol. 57:289–300. Available from: http://www.jstor.org/stable/2346101

Benjamini, Y., and Y. Hochberg. 1995b. Controlling the False Discovery Rate : A Practical and Powerful Approach to Multiple Testing Author (s): Yoav Benjamini and Yosef Hochberg Source : Journal of the Royal Statistical Society . Series B (Methodological), Vol . 57 , No . 1 Published by : J. R. Stat. Soc. Ser. B Stat. Methodol. 57:289–300.

Boettcher, P. J., I. Hoffmann, R. Baumung, A. G. Drucker, C. McManus, P. Berg, A. Stella, L. Nilsen, D. Moran, M. Naves, and M. Thompson. 2014. Genetic resources and genomics for adaptation of livestock to climate change. Front. Genet. 5:2014–2016. doi:10.3389/fgene.2014.00461.

Boettcher, P. J., M. Tixier-Boichard, M. A. Toro, H. Simianer, H. Eding, G. Gandini, S. Joost, D. Garcia, L. Colli, and P. Ajmone-Marsan. 2010. Objectives, criteria and methods for using molecular genetic data in priority setting for conservation of animal genetic resources. Anim. Genet. 41:64–77. doi:10.1111/j.1365-2052.2010.02050.x.

Borg, I., and P. Groenen. 2005. Modern Multidimensional Scaling: Theory and Applications. 2nd ed. Springer Science+Business Media, Inc., New York. Available from: http://www.jstor.org/stable/2669710?origin=crossref

Briggs, H. M., and D. M. Briggs. 1980a. The Angus/Red Angus. In: Modern Breeds of Livestock. 4th ed. Macmillan Publishings Co., Inc., New York. p. 107–128.

Briggs, H. M., and D. M. Briggs. 1980b. The Breeds Developed in the United States. In: Modern Breeds of Livestock. 4th ed. Macmillan Publishings Co., Inc., New York. p. 191–201.

Buchanan, D S, Lenstra, J. A. 2015. Breeds of Cattle. In: D. J. Garrick and A. Ruvinsky, editors. The genetics of cattle. 2nd ed. CAB International. p. 33–66.

Burrow, H. M. 2015. Genetic Aspects of Cattle Adaptation in the Tropics. In: D. J. Garrick and A. Ruvinsky, editors. The Genetics of Cattle. 2nd ed. CAB International, London. p. 571–592.

Cann, H. M., C. de Toma, L. Cazes, M.-F. Legrand, V. Morel, L. Piouffre, J. Bodmer, W. F. Bodmer, B. Bonne-Tamir, A. Cambon-Thomsen, Z. Chen, J. Chu, C. Carcassi, L. Contu, R. Du, L. Excoffier, G. B. Ferrara, J. S. Friedlaender, H. Groot, D. Gurwitz, T. Jenkins, R. J. Herrera, X. Huang, J. Kidd, K. K. Kidd, A. Langaney, A. A. Lin, S. Q. Mehdi, P. Parham, A. Piazza, M. P. Pistillo, Y. Qian, Q. Shu, J. Xu, S. Zhu, J. L. Weber, H. T. Greely, M. W. Feldman, G. Thomas, J. Dausset, and L. L. Cavalli-Sforza. 2002. A Human Genome Diversity Cell Line Panel. Science (80-.). 296:261 LP-262. Available from: http://science.sciencemag.org/content/296/5566/261.2.abstract

Chan, E. K. F., S. H. Nagaraj, and A. Reverter. 2010. The evolution of tropical adaptation: Comparing taurine and zebu cattle. Anim. Genet. 41:467–477. doi:10.1111/j.1365-2052.2010.02053.x.

Chang, C. C., C. C. Chow, L. C. A. M. Tellier, S. Vattikuti, S. M. Purcell, and J. J. Lee. 2015a. Second-generation PLINK : rising to the challenge of larger and richer datasets. 1–16. doi:10.1186/s13742-015-0047-8.

Chang, C. C., C. C. Chow, L. C. Tellier, S. Vattikuti, S. M. Purcell, J. J. Lee, S. Purcell, B. Neale, K. Todd-Brown, L. Thomas, M. Ferreira, D. Bender, B. Browning, S. Browning, B. Howie, P. Donnelly, J. Marchini, A. McKenna, M. Hanna, E. Banks, A. Sivachenko, K. Cibulskis, A. Kernytsky, P. Danecek, A. Auton, G. Abecasis, C. Albers, E. Banks, M. DePristo, H. Li, B. Handsaker, A. Wysoker, T. Fennell, J. Ruan, N. Homer, J. Yang, S. Lee, M. Goddard, P. Visscher, V. Lee, C. Kim, J. Chhugani, M. Deisher, D. Kim, A. Nguyen, I. Haque, V. Pande, W. Walters, H. Hardy, J. Wigginton, D. Cutler, G. Abecasis, S. Guo, E. Thompson, C. Mehta, N. Patel, D. Clarkson, Y. Fan, H. Joe, F. Requena, N. M. Ciudad, S. Lydersen, M. Fagerland, P. Laake, J. Graffelman, V. Moreno, J. Wall, J. Pritchard, S. Gabriel, S. Schaffner, H. Nguyen, J. Moore, J. Roy, B. Blumenstiel, J. Barrett, B. Fry, J. Maller, M. Daly, W. Hill, T. Gaunt, S. Rodríguez, I. Day, D. Taliun, J. Gamper, C. Pattaro, J. Friedman, T. Hastie, H. Höfling, R. Tibshirani, S. Vattikuti, J. Lee, C. Chang, S. Hsu, C. Chow, V. Steiß, T. Letschert, H. Schäfer, R. Pahl, et al. 2015b. Second-generation PLINK: rising to the challenge of larger and richer datasets. Gigascience. 4:7. doi:10.1186/s13742-015-0047-8. Available from: http://www.gigasciencejournal.com/content/4/1/7

Chase, C. C., D. G. Riley, T. A. Olson, S. W. Coleman, A. C. Hammond, C. C. Chase, D. G. Riley, T. A. Olson, S. W. Coleman, and A. C. Hammond. 2004. Maternal and reproductive performance of Brahman x Angus , Senepol x Angus , and Tuli x Angus cows in the subtropics The online version of this article , along with updated information and services , is located on the World Wide Web at : Maternal and repr. J. Anim. Sci. 82:2764–2772.

Collins, M., S.-I. An, W. Cai, A. Ganachaud, E. Guilyardi, F.-F. Jin, M. Jochum, M. Lengaigne, S. Power, A. Timmermann, G. Vecchi, and A. Wittenberg. 2010. The impact of global warming on the tropical Pacific Ocean and El Nino. Nat. Geosci. 3:391–397. doi:10.1038/NGEO868. Available from: http://dx.doi.org/10.1038/ngeo868

Conomos, M. P., M. Miller, and T. Thornton. 2016. Robust Inference of Population Structure for Ancestry Prediction and Correction of Stratification in the Presence of Relatedness. 39:276–293. doi:10.1002/gepi.21896.Robust.

Conomos, M. P., A. P. Reiner, B. S. Weir, and T. A. Thornton. 2016. Model-free Estimation of Recent Genetic Relatedness. Am. J. Hum. Genet. 98:127–148. doi:10.1016/j.ajhg.2015.11.022. Available from: http://dx.doi.org/10.1016/j.ajhg.2015.11.022

Cundiff, L. V., R. M. Thallman, and L. A. Kuehn. 2012. Impact of Bos indicus Genetics on the Global Beef Industry. In: Beef Improvement Federation 44th Annual Research Symposium and Annual Meeting, Houston, TX, April 18, 2012. p. 147–151.

Decker, J., S. McKay, and M. Rolf. 2013. Worldwide Patterns of Ancestry, Divergence, and Admixture in Domesticated Cattle. arXiv Prepr. arXiv …. 10:e1004254. doi:10.1371/journal.pgen.1004254. Available from: http://dx.plos.org/10.1371/journal.pgen.1004254%5Cnhttp://arxiv.org/abs/1309.5118

van Dijk, J., N. D. Sargison, F. Kenyon, and P. J. Skuce. 2010. Climate change and infectious disease: helminthological challenges to farmed ruminants in temperate regions. Animal. 4:377–392. doi:10.1017/s1751731109990991. Available from: http://journals.cambridge.org/download.php?file=/ANM/ANM4_03/S1751731109990991a.pdf&code=531f93558280a66fd69390f92ddf4fcd%5Cnhttp://journals.cambridge.org/download.php?file=/ANM/ANM4_03/S1751731109990991a.pdf&code=6000a198eb60db14e5fb4dfe7e2a40f0

Dikmen, S., J. B. Cole, D. J. Null, and P. J. Hansen. 2013. Genome-Wide Association Mapping for Identification of Quantitative Trait Loci for Rectal Temperature during Heat Stress in Holstein Cattle. 8:1–7. doi:10.1371/journal.pone.0069202.

Dodds, K. G., B. Auvray, S.-A. N. Newman, and J. C. McEwan. 2014. Genomic breed prediction in New Zealand sheep. BMC Genet. 15:92. doi:10.1186/s12863-014-0092-9. Available from: http://www.ncbi.nlm.nih.gov/pubmed/25223795

Drucker, A. G. 2010. Where ’ s the beef ? The economics of AnGR conservation and its in fl uence on policy design and implementation. Anim. Genet. Resour. 47:85–90.

doi:10.1017/S2078633610000913.

Drucker, A. G., V. Gomez, and S. Anderson. 2001. The economic valuation of farm animal genetic resources: A survey of available methods. Ecol. Econ. 36:1–18. doi:10.1016/S0921-8009(00)00242-1.

Ebi, K. L., and K. Bowen. 2016. Extreme events as sources of health vulnerability : Drought as an example. Weather Clim. Extrem. 11:95–102. doi:10.1016/j.wace.2015.10.001. Available from: http://dx.doi.org/10.1016/j.wace.2015.10.001

Ellegren, H., and N. Ellegren. 2016. Determinants of genetic diversity. Nat. Publ. Gr. 17:422–433. doi:10.1038/nrg.2016.58. Available from: http://dx.doi.org/10.1038/nrg.2016.58

Elzo, M. A., and D. L. Wakeman. 1998. Covariance Components and Prediction for Additive and Nonadditive Preweaning Growth Genetic Effects in an Angus-Brahman Multibreed Herd. J. Anim. Sci. 76:1290–1302.

Engelhardt, B. E., and M. Stephens. 2010a. Analysis of population structure: A unifying framework and novel methods based on sparse factor analysis. PLoS Genet. 6. doi:10.1371/journal.pgen.1001117.

Engelhardt, B. E., and M. Stephens. 2010b. Analysis of population structure: A unifying framework and novel methods based on sparse factor analysis. PLoS Genet. 6. doi:10.1371/journal.pgen.1001117.

Fallis, A. . 2012. Environmental Stress and Amelioration in Livestock Production. (V. Sejian, S. M. K. Naqvi, T. Ezeji, J. Lakritz, and R. Lal, editors.). Springer.

FAO. 2007. GLOBAL PLAN OF ACTION FOR ANIMAL GENETIC RESOURCES and the INTERLAKEN DECLARATION. In: World's Poultry Science Journal. Rome, Italy. p. 286. Available from: http://www.journals.cambridge.org/abstract_S0043933909000245

Felius, M., P. A. Koolmees, B. Theunissen, and J. A. Lenstra. 2011. On the breeds of cattle-Historic and current classifications. Diversity. 3:660–692. doi:10.3390/d3040660.

Forster, P., V. Ramaswamy, P. Artaxo, T. Berntsen, R. Betts, D. W. Fahey, J. Haywood, J. Lean, D. C. Lowe, G. Myhre, J. Nganga, R. Prinn, G. Raga, M. S. And, and R. Van Dorland. 2007. Changes in Atmospheric Constituents and in Radiative Forcing. In: Climate Change 2007: The Physical Science Basis. Contribution of Working Group I to the Fourth Assessment Report of the Intergovernmental Panel on Climate Change. (M. T. and H. L. M. Solomon, S., D. Qin, M. Manning, Z. Chen, M. Marquis, K.B. Averyt, editor.). Cambridge University Press, Cambridge, United Kingdom and New York, NY, USA. Available from: http://en.scientificcommons.org/23467316

Francois, O., H. Martins, K. Caye, and S. D. Schoville. 2016. Controlling false discoveries in genome scans for selection. Mol. Ecol. 25:454–469.

doi:10.1111/mec.13513.

Franks, S. J., and A. a. Hoffmann. 2011. Genetics of Climate Change Adaptation. Annu. Rev. Genet. 46:120830114430006. doi:10.1146/annurev-genet-110711-155511.

Franks, S. J., and A. A. Hoffmann. 2012. Genetics of Climate Change Adaptation. Annu. Rev. Genet. 46:185–208. doi:10.1146/annurev-genet-110711-155511.

Frichot, E., and O. Franc. 2015. APPLICATION LEA : An R package for landscape and ecological association studies. 925–929. doi:10.1111/2041-210X.12382.

Frichot, E., O. François, and B. O'Meara. 2015. LEA: AnRpackage for landscape and ecological association studies. Methods Ecol. Evol. 6:925–929. doi:10.1111/2041-210x.12382.

Frichot, E., F. Mathieu, T. Trouillon, G. Bouchard, and O. François. 2014. Fast and efficient estimation of individual ancestry coefficients. Genetics. 196:973–983. doi:10.1534/genetics.113.160572.

Frichot, E., S. D. Schoville, G. Bouchard, and O. Franc. 2013. Testing for Associations between Loci and Environmental Gradients Using Latent Factor Mixed Models. 30:1687–1699. doi:10.1093/molbev/mst063.

Frichot, E., S. D. Schoville, P. De Villemereuil, O. E. Gaggiotti, and O. François. 2015. Detecting adaptive evolution based on association with ecological gradients : Orientation matters ! 22–28. doi:10.1038/hdy.2015.7.

Frichot, R. I. C., R. I. C. Bazin, O. Franc, and P. D. E. Villemereuil. 2014. Genome scan methods against more complex models : when and how much should we trust them ? 2006–2019. doi:10.1111/mec.12705.

Frkonja, A., B. Gredler, U. Schnyder, I. Curik, and J. S. Lkner. 2012a. Prediction of breed composition in an admixed cattle population. Anim. Genet. 43:696–703. doi:10.1111/j.1365-2052.2012.02345.x.

Frkonja, A., B. Gredler, U. Schnyder, I. Curik, and J. Sölkner. 2012b. Prediction of breed composition in an admixed cattle population. Anim. Genet. 43:696–703. doi:10.1111/j.1365-2052.2012.02345.x.

FRKONJA, A., B. GREDLER, U. SCHNYDER, I. CURIK, and J. SÖLKNER. 2011. How to Use Fewer Markers in Admixture Studies ? Agric. Conspec. Sci. cus. 76:187–190.

Frkonja, A., H. W. Raadsma, E. Jonas, G. Thaller, E. Gootwine, E. Seroussi, C. Fuerst, and B. Gredler. 2010. Estimation of individual levels of admixture in crossbred populations from SNP chip data : examples with sheep and cattle populations. Interbull Bull. 62–66.

Funkhouser, S. A., R. O. Bates, C. W. Ernst, D. Newcom, and J. P. Steibel. 2016.

Estimation of genome-wide and locus-specific breed composition in pigs. 1–30. doi:10.2527/tas2016.0003.

Gao, X., and J. Starmer. 2007. Human population structure detection via multilocus genotype clustering. BMC Genet. 8:34. doi:10.1186/1471-2156-8-34. Available from: http://www.ncbi.nlm.nih.gov/pubmed/17592628%5Cnhttp://www.pubmedcentral.nih.gov/articlerender.fcgi?artid=1934381&tool=pmcentrez&rendertype=abstract%5Cnhttp://www.ncbi.nlm.nih.gov/pubmed/17592628%5Cnhttp://www.pubmedcentral.nih.gov/articlerender.fcgi?artid

Garrick, D. J., and A. Ruvinsky, eds. 2015. The Genetics of Cattle. 2nd ed. CAB International, Boston.

Gaughan, J. B., T. L. Mader, S. M. Holt, M. L. Sullivan, and G. L. Hahn. 2010. Assessing the heat tolerance of 17 beef cattle genotypes. Int. J. Biometeorol. 54:617–627. doi:10.1007/s00484-009-0233-4.

Gould, J. 2015. Core facilities: Shared support. Nature. 519:495–496. doi:doi:10.1038/nj7544-495a.

Grey, C. 1919. WITH A GLANCE BACK AS WE STEP UP AND INTO A NEW ERA. Aberdeen-Angus J. 1:8.

Groeneveld, L. F., J. A. Lenstra, H. Eding, M. A. Toro, B. Scherf, D. Pilling, R. Negrini, E. K. Finlay, H. Jianlin, E. Groeneveld, and S. Weigend. 2010. Genetic diversity in farm animals - A review. Anim. Genet. 41:6–31. doi:10.1111/j.1365-2052.2010.02038.x.

Harris, M. A., J. I. Deegan, A. Ireland, J. Lomax, M. Ashburner, S. Tweedie, S. Carbon, S. Lewis, C. Mungall, J. Day-Richter, K. Eilbeck, J. A. Blake, C. Bult, A. D. Diehl, M. Dolan, H. Drabkin, J. T. Eppig, D. P. Hill, L. Ni, M. Ringwald, R. Balakrishnan, G. Binkley, J. M. Cherry, K. R. Christie, M. C. Costanzo, Q. Dong, S. R. Engel, D. G. Fisk, J. E. Hirschman, B. C. Hitz, E. L. Hong, C. J. Krieger, S. R. Miyasato, R. S. Nash, J. Park, M. S. Skrzypek, S. Weng, E. D. Wong, K. K. Zhu, D. Botstein, K. Dolinski, M. S. Livstone, R. Oughtred, T. Berardini, D. Li, S. Y. Rhee, R. Apweiler, D. Barrell, E. Camon, E. Dimmer, R. Huntley, N. Mulder, V. K. Khodiyar, R. C. Lovering, S. Povey, R. Chisholm, P. Fey, P. Gaudet, W. Kibbe, R. Kishore, E. M. Schwarz, P. Sternberg, K. Van Auken, M. G. Giglio, L. Hannick, J. Wortman, M. Aslett, M. Berriman, V. Wood, H. Jacob, S. Lauledérkind, V. Petri, M. Shimoyama, J. Smith, S. Twigger, P. Jaiswal, T. Seigfried, D. Howe, M. Westerfield, C. Collmer, T. Torto-Alalibo, E. Feltrin, G. Valle, S. Bromberg, S. Burgess, and F. McCarthy. 2008. The Gene Ontology project in 2008. Nucleic Acids Res. 36:440–444. doi:10.1093/nar/gkm883.

Hayes, B. J., P. J. Bowman, A. J. Chamberlain, K. Savin, C. P. van Tassell, T. S. Sonstegard, and M. E. Goddard. 2009. A validated genome wide association study to breed cattle adapted to an environment altered by climate change. PLoS One. 4:1–8. doi:10.1371/journal.pone.0006676.

Hayes, B. J., H. A. Lewin, and M. E. Goddard. 2013. The future of livestock breeding:

Genomic selection for efficiency, reduced emissions intensity, and adaptation. Trends Genet. 29:206–214. doi:10.1016/j.tig.2012.11.009. Available from: http://dx.doi.org/10.1016/j.tig.2012.11.009

Hijmans, R. J. 2016. geosphere: Spherical Trigonometry. Available from: https://cran.r-project.org/package=geosphere

Hill, W. G., and X.-S. Zhang. 2009. Maintaining Genetic Variation in Fitness. In: J. Van der Werf, H. U. Graser, R. Frankham, and C. Gondro, editors. Adaptation and Fitness in Animal Populations. Springer. p. 59–81. Available from: http://dx.doi.org/10.1007/978-1-4020-9005-9_5

Hoffmann, A., P. Griffin, S. Dillon, R. Catullo, R. Rane, M. Byrne, R. Jordan, J. Oakeshott, A. Weeks, L. Joseph, P. Lockhart, J. Borevitz, and C. Sgrò. 2015. A framework for incorporating evolutionary genomics into biodiversity conservation and management. Clim. Chang. Responses. 2:1. doi:10.1186/s40665-014-0009-x. Available from: http://climatechangeresponses.biomedcentral.com/articles/

Hoffmann, I. 2010. Climate change and the characterization, breeding and conservation of animal genetic resources. Anim. Genet. 41:32–46. doi:10.1111/j.1365-2052.2010.02043.x.

Hoffmann, I. 2013. Adaptation to climate change--exploring the potential of locally adapted breeds. Animal. 7:346–362. doi:10.1017/S1751731113000815. Available from: http://www.ncbi.nlm.nih.gov/pubmed/23739476

Holderegger, R., U. Kamm, and F. Gugerli. 2006. Adaptive vs. neutral genetic diversity: Implications for landscape genetics. Landsc. Ecol. 21:797–807. doi:10.1007/s10980-005-5245-9.

Illumna. 2015. OvineSNP50 Genotyping BeadChip.

Ingram, K. T., K. Dow, L. Carter, and J. (Eds. . Anderson. 2013. Climate of the Southeast United States: Variability, Change, Impacts, and Vulnerability.

De Iorio, M., L. T. Elliott, S. Favaro, K. Adhikari, and Y. W. Teh. 2015. Modeling population structure under hierarchical Dirichlet processes. 1–26. Available from: http://arxiv.org/abs/1503.08278

Jolliffe, I. T. 2002. Principal Component Analysis. 2nd ed. Springer. Available from: http://onlinelibrary.wiley.com/doi/10.1002/0470013192.bsa501/full

Joost, S., a. Bonin, M. W. Bruford, L. Després, C. Conord, G. Erhardt, and P. Taberlet. 2007. A spatial analysis method (SAM) to detect candidate loci for selection: Towards a landscape genomics approach to adaptation. Mol. Ecol. 16:3955–3969. doi:10.1111/j.1365-294X.2007.03442.x.

Joost, S., S. Vuilleumier, J. D. Jensen, S. Schoville, K. Leempoel, S. Stucki, I. Widmer,

C. Melodelima, J. Rolland, and S. Manel. 2013. Uncovering the genetic basis of adaptive change: On the intersection of landscape genomics and theoretical population genetics. Mol. Ecol. 22:3659–3665. doi:10.1111/mec.12352.

Kijas, J. W., J. A. Lenstra, B. Hayes, S. Boitard, L. R. Neto, M. S. Cristobal, B. Servin, R. McCulloch, V. Whan, K. Gietzen, S. Paiva, W. Barendse, E. Ciani, H. Raadsma, J. McEwan, and B. Dalrymple. 2012. Genome-wide analysis of the world's sheep breeds reveals high levels of historic mixture and strong recent selection. PLoS Biol. 10. doi:10.1371/journal.pbio.1001258.

Kijas, J. W., and A. K. Naumova. 2014. Haplotype-based analysis of selective sweeps in sheep. Genome. 57:433–437. doi:10.1139/gen-2014-0049. Available from: http://www.nrcresearchpress.com/doi/abs/10.1139/gen-2014-0049

Kim, E.-S., A. R. Elbeltagy, A. M. Aboul-Naga, B. Rischkowsky, B. Sayre, J. M. Mwacharo, and M. F. Rothschild. 2015. Multiple genomic signatures of selection in goats and sheep indigenous to a hot arid environment. Heredity (Edinb). 116:1–10. doi:10.1038/hdy.2015.94. Available from: http://www.nature.com/doifinder/10.1038/hdy.2015.94

Kim, E. S., and M. F. Rothschild. 2014. Genomic adaptation of admixed dairy cattle in East Africa. Front. Genet. 5:1–10. doi:10.3389/fgene.2014.00443.

Komender, P. 1988. Crossbreeding in farm animals. J. Anim. Breed. Genet. 105:362–371. doi:10.1111/j.1439-0388.1988.tb00308.x. Available from: http://doi.wiley.com/10.1111/j.1439-0388.1988.tb00308.x

Kuehn, L. A., J. W. Keele, G. L. Bennett, T. G. Mcdaneld, T. P. L. Smith, W. M. Snelling, T. S. Sonstegard, and R. M. Thallman. 2011. Predicting breed composition using breed frequencies of 50,000 markers from the US Meat Animal Research Center 2,000 Bull Project. 1742–1750. doi:10.2527/jas.2010-3530.

Kuehn, L. A., J. W. Keele, G. L. Bennett, T. G. McDaneld, T. P. L. Smith, W. M. Snelling, T. S. Sonstegard, and R. M. Thallman. 2011. Predicting breed composition using breed frequencies of 50,000 markers from the US Meat Animal Research Center 2,000 bull project. J. Anim. Sci. 89:1742–1750. doi:10.2527/jas.2010-3530.

Lachance, J., and S. A. Tishkoff. 2013. SNP ascertainment bias in population genetic analyses: Why it is important, and how to correct it. BioEssays. 35:780–786. doi:10.1002/bies.201300014.

Lawson, D. J., and D. Falush. 2012. Similarity matrices and clustering algorithms for population identification using genetic data. Annu. Rev. Genomics Hum. Genet. 1–11. doi:10.1146/annurev-genom-082410-101510. Available from: http://www.annualreviews.org/doi/pdf/10.1146/annurev-genom-082410-101510

Lawson, D. J., G. Hellenthal, S. Myers, and D. Falush. 2012. Inference of population structure using dense haplotype data. PLoS Genet. 8:11–17.

doi:10.1371/journal.pgen.1002453.

LAWSON, M. P., and C. W. STOCKTON. 1981. Desert Myth and Climatic Reality. Ann. Assoc. Am. Geogr. 71:527–535. doi:10.1111/j.1467-8306.1981.tb01372.x. Available from: http://www.tandfonline.com/doi/abs/10.1111/j.1467-8306.1981.tb01372.x

Lenstra, J. A., and M. Felius. 2015. Genetic Aspects of Domestication. In: D. J. Garrick and A. Ruvinsky, editors. The Genetics of Cattle. 2nd ed. CAB International, Boston. p. 19–28.

Lenstra, J. A., M. Felius, and B. Theunissen. 2014. Domestic cattle and buffaloes. In: M. Melletti and J. Burton, editors. Ecology, Evolution and Behaviour of Wild Cattle: Implications for Conservation. Cambridge University Press. p. 30–38.

Lewis, J., Z. Abas, C. Dadousis, D. Lykidis, and P. Paschou. 2011. Tracing Cattle Breeds with Principal Components Analysis Ancestry Informative SNPs. 6. doi:10.1371/journal.pone.0018007.

Littell, J.S., M. M. Elsner, L. C. W. Binder, and A. . (eds) Snover. 2009. The Washington Climate Change Impacts Assessment Evaluating Washington's Future.

Liu, N., and H. Zhao. 2006. A non-parametric approach to population structure inference using multilocus genotypes. Hum. Genomics. 2:353. doi:10.1186/1479-7364-2-6-353. Available from: http://www.humgenomics.com/content/2/6/353

Liu, Y., T. Nyunoya, S. Leng, S. A. Belinsky, Y. Tesfaigzi, and S. Bruse. 2013. Softwares and methods for estimating genetic ancestry in human populations. Hum Genomics. 7:1. doi:10.1186/1479-7364-7-1.

Long, J. C. 1991. The genetic structure of admixed populations. Genetics. 127:417–428.

Lowry, D. B. 2010. Landscape evolutionary genomics. Biol. Lett. 6:502–504. doi:10.1098/rsbl.2009.0969.

Luikart, G., P. R. England, D. Tallmon, S. Jordan, and P. Taberlet. 2003. The power and promise of population genomics: from genotyping to genome typing. Nat. Rev. Genet. 4:981–994. doi:10.1038/nrg1226.

Lv, F., S. Agha, J. Kantanen, L. Colli, S. Stucki, J. W. Kijas, M. Li, and P. A. Marsan. 2014. Adaptations to Climate-Mediated Selective Pressures in Sheep. 31:3324–3343. doi:10.1093/molbev/msu264.

MacDonald, J., and J. Sinclair. 1910. History of Aberdeen-Angus Cattle. Vinton & Company, Ltd., London.

Manel, S., and R. Holderegger. 2013. Ten years of landscape genetics. Trends Ecol. Evol. 28:614–621. doi:10.1016/j.tree.2013.05.012.

Manel, S., S. Joost, B. K. Epperson, R. Holderegger, A. Storfer, M. S. Rosenberg, K. T. Scribner, A. Bonin, and M. J. Fortin. 2010. Perspectives on the use of landscape genetics to detect genetic adaptive variation in the field. Mol. Ecol. 19:3760–3772. doi:10.1111/j.1365-294X.2010.04717.x.

Manichaikul, A., J. C. Mychaleckyj, S. S. Rich, K. Daly, M. Sale, and W. M. Chen. 2010. Robust relationship inference in genome-wide association studies. Bioinformatics. 26:2867–2873. doi:10.1093/bioinformatics/btq559.

Matuszewski, S., J. Hermisson, and M. Kopp. 2015. Catch me if you can: Adaptation from standing genetic variation to a moving phenotypic optimum. Genetics. 200:1255–1274. doi:10.1534/genetics.115.178574.

Maudet, C., G. Luikart, and P. Taberlet. 2002. Genetic diversity and assignment tests among seven French cattle breeds based on microsatellite DNA analysis. J. Anim. Sci. 80:942–950.

McLaren, W., L. Gil, S. E. Hunt, H. S. Riat, G. R. S. Ritchie, A. Thormann, P. Flicek, and F. Cunningham. 2016. The Ensembl Variant Effect Predictor. bioRxiv. 42374. doi:10.1101/042374. Available from: http://biorxiv.org/content/early/2016/03/04/042374.abstract

McNeill, A. B., ed. 2010. Climate Change and Its Causes, Effects and Prediction: Assessing Climate Change Impacts on the United States. Nova Science Publishers, Inc., New York.

McVean, G. 2009. A genealogical interpretation of principal components analysis. PLoS Genet. 5. doi:10.1371/journal.pgen.1000686.

Meinshausen, M., S. J. Smith, K. Calvin, J. S. Daniel, M. L. T. Kainuma, J. Lamarque, K. Matsumoto, S. A. Montzka, S. C. B. Raper, K. Riahi, A. Thomson, G. J. M. Velders, and D. P. P. van Vuuren. 2011. The RCP greenhouse gas concentrations and their extensions from 1765 to 2300. Clim. Change. 109:213–241. doi:10.1007/s10584-011-0156-z.

Messer, P. W., and D. A. Petrov. 2013. Population genomics of rapid adaptation by soft selective sweeps. Trends Ecol. Evol. 28:659–669. doi:10.1016/j.tree.2013.08.003. Available from: http://dx.doi.org/10.1016/j.tree.2013.08.003

Mi, H., X. Huang, A. Muruganujan, H. Tang, C. Mills, D. Kang, and P. D. Thomas. 2017. PANTHER version 11: expanded annotation data from Gene Ontology and Reactome pathways, and data analysis tool enhancements. Nucleic Acids Res. 45:D183–D189. doi:10.1093/nar/gkw1138. Available from: https://academic.oup.com/nar/article-lookup/doi/10.1093/nar/gkw1138

Mi, H., A. Muruganujan, J. T. Casagrande, and P. D. Thomas. 2013. Large-scale gene function analysis with the PANTHER classification system. Nat. Protoc. 8:1551–1566. doi:10.1038/nprot.2013.092.

De Mita, S., A. C. Thuillet, L. Gay, N. Ahmadi, S. Manel, J. Ronfort, and Y. Vigouroux. 2013. Detecting selection along environmental gradients: Analysis of eight methods and their effectiveness for outbreeding and selfing populations. Mol. Ecol. 22:1383–1399. doi:10.1111/mec.12182.

Morota, G., F. Peñagaricano, J. L. Petersen, D. C. Ciobanu, K. Tsuyuzaki, and I. Nikaido. 2015. An application of MeSH enrichment analysis in livestock. Anim. Genet. 46:381–387. doi:10.1111/age.12307.

Moss, R. H., J. A. Edmonds, K. A. Hibbard, M. R. Manning, S. K. Rose, D. P. Van Vuuren, T. R. Carter, S. Emori, M. Kainuma, T. Kram, G. A. Meehl, J. F. B. Mitchell, N. Nakicenovic, K. Riahi, S. J. Smith, R. J. Stouffer, A. M. Thomson, J. P. Weyant, and T. J. Wilbanks. 2010. The next generation of scenarios for climate change research and assessment. Nature. 463:747–756. doi:10.1038/nature08823. Available from: http://www.scopus.com/record/display.url?eid=2-s2.0-76749096338&origin=inward&txGid=CwkSAJPyATm2B6yoCS28rHm:11

Nardone, A., B. Ronchi, N. Lacetera, M. S. Ranieri, and U. Bernabucci. 2010. Effects of climate changes on animal production and sustainability of livestock systems. Livest. Sci. 130:57–69. doi:10.1016/j.livsci.2010.02.011.

Naskar, S., G. R. Gowane, A. Chopra, C. Paswan, and L. L. L. Prince. 2012. Genetic Adaptability of Livestock to Environmental Stresses. In: V. Sejian, S. M. K. Naqvi, T. Ezeji, J. Lakritz, and R. Lal, editors. Environmental Stress and Amelioration in Livestock Production. Springer, Berlin Heidelberg. p. 319–367.

Nei, M., Y. Suzuki, and M. Nozawa. 2010. The Neutral Theory of Molecular Evolution in the Genomic Era. Annu. Rev. Genomics Hum. Genet. 11:265–89. doi:10.1146/annurev-genom-082908-150129.

NEOGEN. 2017. GGP F-250 for Beef. Available from: http://genomics.neogen.com/en/ggp-f-250-beef

Nienaber, J. A., G. L. Hahn, T. M. Brown-Brandl, and R. A. Eigenberg. 2007. Summer heat waves - extreme years. Available from: http://www.scopus.com/inward/record.url?eid=2-s2.0-35648964079&partnerID=40&md5=b15bd4130d4263951a4b65b9bc68e4a8

NOAA. 2013. Regional Climate Trends and Scenarios for the U.S. National Climate Assessment: Climate of the Contiguous United States.

NOAA. 2017. Global Surface Summary of the Day - GSOD - NOAA Data Catalog. Available from: https://data.noaa.gov/dataset/global-surface-summary-of-the-day-gsod

Pachauri, R. K., and L. A. Meyer. 2014. IPCC, 2014: Climate Change 2014: Synthesis Report. Contribution of Working Groups I, II and III to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change.

Padhukasahasram, B. 2014. Inferring ancestry from population genomic data and its applications. Front. Genet. 5:1–5. doi:10.3389/fgene.2014.00204.

Parry, M. L., C. Rosenzweig, A. Iglesias, M. Livermore, and G. Fischer. 2004. Effects of climate change on global food production under SRES emissions and socio-economic scenarios. Glob. Environ. Chang. 14:53–67. doi:10.1016/j.gloenvcha.2003.10.008.

Paschou, P., P. Drineas, J. Lewis, C. M. Nievergelt, D. A. Nickerson, J. D. Smith, P. M. Ridker, D. I. Chasman, R. M. Krauss, and E. Ziv. 2008. Tracing Sub-Structure in the European American Population with PCA-Informative Markers. PLoS Genet. 4. doi:10.1371/journal.pgen.1000114.

Paschou, P., J. Lewis, A. Javed, and P. Drineas. 2010a. Ancestry informative markers for fine-scale individual assignment to worldwide populations. J. Med. Genet. 47:835–847. doi:10.1136/jmg.2010.078212.

Paschou, P., J. Lewis, A. Javed, and P. Drineas. 2010b. Ancestry informative markers for fine-scale individual assignment to worldwide populations. J. Med. Genet. 47:835–847. doi:10.1136/jmg.2010.078212.

Paschou, P., E. Ziv, E. G. Burchard, S. Choudhry, W. Rodriguez-cintron, M. W. Mahoney, and P. Drineas. 2007. PCA-Correlated SNPs for Structure Identification in Worldwide Human Populations. PLoS Genet. 3. doi:10.1371/journal.pgen.0030160.

Patterson, N., A. L. Price, and D. Reich. 2006. Population structure and eigenanalysis. PLoS Genet. 2:2074–2093. doi:10.1371/journal.pgen.0020190.

Pfaff, C. L., E. J. Parra, C. Bonilla, K. Hiester, P. M. McKeigue, M. I. Kamboh, R. G. Hutchinson, R. E. Ferrell, E. Boerwinkle, and M. D. Shriver. 2001. Population structure in admixed populations: effect of admixture dynamics on the pattern of linkage disequilibrium. Am. J. Hum. Genet. 68:198–207. doi:10.1086/316935. Available from: http://www.pubmedcentral.nih.gov/articlerender.fcgi?artid=1234913&tool=pmcentrez&rendertype=abstract

Porras-Hurtado, L., Y. Ruiz, C. Santos, C. Phillips, Á. Carracedo, and M. V. Lareu. 2013. An overview of STRUCTURE: Applications, parameter settings, and supporting software. Front. Genet. 4:1–13. doi:10.3389/fgene.2013.00098.

Price, A. L., N. a Zaitlen, D. Reich, and N. Patterson. 2010. New approaches to population stratification in genome-wide association studies. Nat. Rev. Genet. 11:459–463. doi:10.1038/nrg2813. Available from: http://dx.doi.org/10.1038/nrg2813

Pritchard, J. K., and N. a Rosenberg. 1999. Use of unlinked genetic markers to detect population stratification in association studies. Am. J. Hum. Genet. 65:220–8. doi:10.1086/302449. Available from: http://www.ncbi.nlm.nih.gov/pubmed/10364535

Pritchard, J. K., M. Stephens, and P. Donnelly. 2000. Inference of population structure using multilocus genotype data. Genetics. 155:945–959.

Proudfoot, C., D. F. Carlson, R. Huddart, C. R. Long, D. G. Mclaren, C. B. A. Whitelaw, and S. C. Fahrenkrug. 2015. Genome edited sheep and cattle. 147–153. doi:10.1007/s11248-014-9832-x.

QIAGEN. 2016. QIAamp® DNA Mini and Blood Mini Handbook. Available from: http://www.qiagen.com/knowledge-and-support/resource-center/resource-download.aspx?id=67893a91-946f-49b5-8033-394fa5d752ea&lang=en

QUAGEN. 2006. DNeasy® Blood & Tissue Handbook.

R Core Team. 2013. R: A language and environment for statistical computing. R Found. Stat. Comput. Available from: http://www.r-project.org/

R Core Team. 2016. R: A language and environment for statistical computing. Available from: https://www.r-project.org/

Raj, A., M. Stephens, and J. K. Pritchard. 2014. FastSTRUCTURE: Variational inference of population structure in large SNP data sets. Genetics. 197:573–589. doi:10.1534/genetics.114.164350.

Ramey, H. R., J. E. Decker, S. D. McKay, M. M. Rolf, R. D. Schnabel, and J. F. Taylor. 2013. Detection of selective sweeps in cattle using genome-wide SNP data. BMC Genomics. 14:382. doi:10.1186/1471-2164-14-382. Available from: http://bmcgenomics.biomedcentral.com/articles/10.1186/1471-2164-14-382

Reilly, J., F. Tubiello, B. McCarl, D. Abler, R. Darwin, K. Fuglie, S. Hollinger, C. Izaurralde, S. Jagtap, and J. Jones. 2003. US agriculture and climate change: new results. Clim. Change. 57:43–67. doi:10.1023/A:1022103315424. Available from: http://link.springer.com/article/10.1023/A:1022103315424

Rellstab, C., F. Gugerli, A. J. Eckert, A. M. Hancock, and R. Holderegger. 2015. A practical guide to environmental association analysis in landscape genomics. Mol. Ecol. 24:4348–4370. doi:10.1111/mec.13322.

Renaudeau, D., A. Collin, S. Yahav, V. de Basilio, J. L. Gourdine, and R. J. Collier. 2012. Adaptation to hot climate and strategies to alleviate heat stress in livestock production. Animal. 6:707–728. doi:10.1017/s1751731111002448.

Riley, D. G., C. C. Chase, S. W. Coleman, and T. A. Olson. 2007. Evaluation of birth and weaning traits of Romosinuano calves as purebreds and crosses with Brahman and Angus. J. Anim. Sci. 85:289–298. doi:10.2527/jas.2006-416.

Rosenberg, N. a. 2002. Genetic Structure of Human Populations. Science (80-.). 238:2381–2386. doi:10.1126/science.1078311.

Rosenberg, N. A., L. M. Li, R. Ward, and J. K. Pritchard. 2003. Informativeness of Genetic Markers for Inference of Ancestry. Am. J. Hum. Genet. 73:1402–1422. doi:10.1086/380416.

Roughgarden, J. 1979. Theory of Population Genetics and Evolutionary Ecology: An Introduction. Macmillan Publishings Co., Inc., New York.

Salinger, J., M. V. K. Sivakumar, and R. P. Motha, eds. 2005. Increasing Climate Variability: Reducing the Vulnerability of Agriculture and Forestry. Springer, Dordrecht, The Netherlands.

Schmidhuber, J., and F. N. Tubiello. 2007. Global food security under climate change. Proc. Natl. Acad. Sci. U. S. A. 104:19703–8. doi:10.1073/pnas.0701976104. Available from: http://www.pnas.org/content/104/50/19703.full

Schnabel, R., E. Simpson, D. Larkin, J. Hoff, J. Decker, and J. Taylor. 2016. Design and Application of the Cattle GGP-F250 Assay. Anim. Genet. Prep.

Sheets, E. W. 1915. Breeds of Cattle. US Dep. Agric. Farmers Bull. 1–22.

Shringarpure, S. S., C. D. Bustamante, K. L. Lange, and D. H. Alexander. 2016. Efficient analysis of large datasets and sex bias with ADMIXTURE. bioarXiv. 1:1–10. doi:10.1101/039347. Available from: http://biorxiv.org/biorxiv/early/2016/02/10/039347.full.pdf

Shringarpure, S., and E. P. Xing. 2014. Effects of Sample Selection Bias on the Accuracy of Population Structure and Ancestry Inference. G3 Genes|Genomes|Genetics. 4:901–911. doi:10.1534/g3.113.007633. Available from: http://www.g3journal.org/content/4/5/901%5Cnhttp://www.ncbi.nlm.nih.gov/pubmed/24637351

Sinnock, P. 1975. The Wahlund Effect for the Two-Locus Model. Am. Nat. Nat. 109:565–570.

Sinnott, R. W. 1984. Virtues of the Haversine. Sky Telescope. 68:159.

Smedley, D., S. Haider, S. Durinck, L. Pandini, P. Provero, J. Allen, O. Arnaiz, M. H. Awedh, R. Baldock, G. Barbiera, P. Bardou, T. Beck, A. Blake, M. Bonierbale, A. J. Brookes, G. Bucci, I. Buetti, S. Burge, C. Cabau, J. W. Carlson, C. Chelala, C. Chrysostomou, D. Cittaro, O. Collin, R. Cordova, R. J. Cutts, E. Dassi, A. Di Genova, A. Djari, A. Esposito, H. Estrella, E. Eyras, J. Fernandez-Banet, S. Forbes, R. C. Free, T. Fujisawa, E. Gadaleta, J. M. Garcia-Manteiga, D. Goodstein, K. Gray, J. A. Guerra-Assunção, B. Haggarty, D.-J. Han, B. W. Han, T. Harris, J. Harshbarger, R. K. Hastings, R. D. Hayes, C. Hoede, S. Hu, Z.-L. Hu, L. Hutchins, Z. Kan, H. Kawaji, A. Keliet, A. Kerhornou, S. Kim, R. Kinsella, C. Klopp, L. Kong, D. Lawson, D. Lazarevic, J.-H. Lee, T. Letellier, C.-Y. Li, P. Lio, C.-J. Liu, J. Luo, A. Maass, J. Mariette, T. Maurel, S. Merella, A. M. Mohamed, F. Moreews, I. Nabihoudine, N. Ndegwa, C. Noirot, C. Perez-Llamas, M. Primig, A. Quattrone, H. Quesneville, D. Rambaldi, J. Reecy, M. Riba, S. Rosanoff, A. A. Saddiq, E. Salas, O. Sallou, R. Shepherd, R. Simon, L. Sperling, W. Spooner, D. M. Staines, D. Steinbach, K. Stone, E. Stupka, J. W. Teague, A. Z. Dayem Ullah, et al. 2015. The BioMart community portal: an innovative alternative to large, centralized data repositories. Nucleic Acids Res. 43:W589–W598.

doi:10.1093/nar/gkv350. Available from: https://academic.oup.com/nar/article-lookup/doi/10.1093/nar/gkv350

Somero, G. N. 2010. The physiology of climate change: how potentials for acclimatization and genetic adaptation will determine “winners” and “losers.” J. Exp. Biol. 213:912–920. doi:10.1242/jeb.037473.

St-Pierre, N. R., B. Cobanov, and G. Schnitkey. 2003. Economic losses from heat stress by US livestock industries. J. Dairy Sci. 86:E52–E77. doi:10.3168/jds.S0022-0302(03)74040-5. Available from: http://dx.doi.org/10.3168/jds.S0022-0302(03)74040-5

Stapley, J., J. Reger, P. G. D. Feulner, C. Smadja, J. Galindo, R. Ekblom, C. Bennison, A. D. Ball, A. P. Beckerman, and J. Slate. 2010. Adaptation genomics: The next generation. Trends Ecol. Evol. 25:705–712. doi:10.1016/j.tree.2010.09.002. Available from: http://dx.doi.org/10.1016/j.tree.2010.09.002

Tan, W. (Spring), D. F. Carlson, M. W. Walton, S. C. Fahrenkrug, and P. B. Hackett. 2013. Precision Editing of Large Animal Genomes. Adv. Genet. 80:37–97. doi:10.1016/B978-0-12-404742-6.00002-8.Precision.

Tang, H., J. Peng, P. Wang, and N. J. Risch. 2005. Estimation of individual admixture: Analytical and study design considerations. Genet. Epidemiol. 28:289–301. doi:10.1002/gepi.20064.

Thornton, P. K., J. van de Steeg, A. Notenbaert, and M. Herrero. 2009. The impacts of climate change on livestock and livestock systems in developing countries: A review of what we know and what we need to know. Agric. Syst. 101:113–127. doi:10.1016/j.agsy.2009.05.002.

Thornton, T., M. P. Conomos, S. Sverdlov, E. M. Blue, C. Y. Cheung, C. G. Glazner, S. M. Lewis, and E. M. Wijsman. 2014. Estimating and adjusting for ancestry admixture in statistical methods for relatedness inference, heritability estimation, and association testing. BMC Proc. 8:S5. doi:10.1186/1753-6561-8-S1-S5. Available from: http://www.ncbi.nlm.nih.gov/pmc/articles/PMC4143704/%5Cnhttp://www.ncbi.nlm.nih.gov/pmc/articles/PMC4143704/pdf/1753-6561-8-S1-S5.pdf

Thornton, T., H. Tang, T. J. Hoffmann, H. M. Ochs-Balcom, B. J. Caan, and N. Risch. 2012. Estimating kinship in admixed populations. Am. J. Hum. Genet. 91:122–138. doi:10.1016/j.ajhg.2012.05.024.

Turner, J. W. 1980. Genetic and Biological Aspects of Zebu Adaptability. J. Anim. Sci. 50:1201–1205.

Turner, S. 2017. qqman: Q-Q and Manhattan Plots for GWAS Data. Available from: https://cran.r-project.org/package=qqman

Turner, S., L. L. Armstrong, Y. Bradford, C. S. Carlson, C. Dana, A. T. Crenshaw, M. De Andrade, K. F. Doheny, L. Jonathan, G. Hayes, G. Jarvik, L. Jiang, I. J. Kullo, R. Li, T. a

Manolio, M. Matsumoto, C. a Mccarty, N. Andrew, D. B. Mirel, J. E. Paschall, E. W. Pugh, V. Luke, R. a Wilke, R. L. Zuvich, and M. D. Ritchie. 2011. Quality control procedures for genome wide association studies. Curr. Proc. Hum. Genet. 68:1–24. doi:10.1002/0471142905.hg0119s68.Quality.

UNEP. 1998. Handbook on Methods for Climate Change Impact Assessment and Adaptation Strategies. (J. F. Feenstra, I. Burton, J. B. Smith, and R. S. J. Tol, editors.). Available from: http://research.fit.edu/sealevelriselibrary/documents/doc_mgr/465/Global_Methods_for_CC_Assessment_Adaptation_-_UNEP_1998.pdf

Vanraden, P. M., and T. A. Cooper. 2015. Genomic evaluations and breed composition for crossbred U . S . dairy cattle. 2015:1–21.

Verhoeven, K. J. F., M. Macel, L. M. Wolfe, and A. Biere. 2011. Population admixture, biological invasions and the balance between local adaptation and inbreeding depression. Proc. Biol. Sci. 278:2–8. doi:10.1098/rspb.2010.1272.

Vilà, C., J. Seddon, and H. Ellegren. 2005. Genes of domestic mammals augmented by backcrossing with wild ancestors. Trends Genet. 21:214–218. doi:10.1016/j.tig.2005.02.004.

Walthall, C. L., J. Hatfield, P. Backlund, L. Lengnick, E. Marshall, M. Walsh, S. Adkins, M. Aillery, E. A. Ainsworth, C. Ammann, C. J. Anderson, I. Bartomeus, L. H. Baumgard, F. Booker, B. Bradley, D. M. Blumenthal, J. Bunce, K. Burkey, S. M. Dabney, J. A. Delgado, J. Dukes, A. Funk, K. Garrett, M. Glenn, D. A. Grantz, D. Goodrich, S. Hu, R. C. Izaurralde, R. A. C. Jones, S.-H. Kim, A. D. B. Leaky, K. Lewers, T. L. Mader, A. McClung, J. Morgan, D. J. Muth, M. Nearing, D. M. Oosterhuis, D. Ort, C. Parmesan, W. T. Pettigrew, W. Polley, R. Rader, C. Rice, M. Rivington, E. Rosskopf, W. A. Salas, L. E. Sollenberger, R. Srygley, C. Stöckle, E. S. Takle, D. Timlin, J. W. White, R. Winfree, L. Wright-Morton, and L. H. Ziska. 2012. Climate Change and Agriculture in the United States: Effects and Adaptation. Washington, DC. Available from: http://www.usda.gov/oce/climate_change/effects.htm

Weir, B. S., and C. C. Cockerham. 1984. Estimating F-Statistics for the Analysis of Population Structure. Evolution (N. Y). 38:1358–1370. doi:10.2307/2408641.

Weir, B. S., and J. Goudet. 2016. A unified characterization of population structure and relatedness. bioRxiv. Available from: http://biorxiv.org/content/early/2016/11/17/088260.abstract

Wickham, H. 2009. ggplot2: Elegant Graphics for Data Analysis.

Wiggans, G. R., P. M. VanRaden, and T. A. Cooper. 2011. The genomic evaluation system in the United States: Past, present, future. J. Dairy Sci. 94:3202–3211. doi:10.3168/jds.2010-3866. Available from: http://www.ncbi.nlm.nih.gov/pubmed/21605789%5Cnhttp://www.journalofdairyscience.org/article/S0022030211003079/fulltext%5Cnhttp://linkinghub.elsevier.com/retrieve/pii/S0

022030211003079

Wilkinson, S. 2012. Genetic diversity and structure of livestock breeds. Available from: http://www.era.lib.ed.ac.uk/handle/1842/6488

Wilkinson, S., P. Wiener, A. L. Archibald, A. Law, R. D. R. Schnabel, S. D. S. McKay, J. F. J. Taylor, and R. Ogden. 2011a. Evaluation of approaches for identifying population informative markers from high density SNP chips. BMC Genet. 12:45. doi:10.1186/1471-2156-12-45. Available from: http://bmcgenet.biomedcentral.com/articles/10.1186/1471-2156-12-45%0Ahttp://www.pubmedcentral.nih.gov/articlerender.fcgi?artid=3118130&tool=pmcentrez&rendertype=abstract

Wilkinson, S., P. Wiener, A. L. Archibald, A. Law, R. D. R. Schnabel, S. D. S. McKay, J. F. J. Taylor, R. Ogden, P. Waser, C. Strobeck, N. Davies, F. Villablanca, G. Roderick, S. Manel, O. Gaggiotti, R. Waples, S. Roques, P. Duchesne, L. Bernatchez, R. Ciampolini, V. Cetica, E. Ciani, E. Mazzanti, X. Fosella, F. Marroni, M. Biagetti, C. Sebastiani, P. Papa, G. Filippini, B. Rannala, J. Mountain, D. Paetkau, W. Calvert, I. Stirling, C. Strobeck, C. Maudet, G. Luikart, P. Taberlet, R. Negrini, L. Nicoloso, P. Crepaldi, E. Milanesi, L. Colli, F. Chegdani, L. Pariset, S. Dunner, H. Leveziel, J. Williams, P. Morin, G. Luikart, R. Wayne, S. Grp, S. Kim, A. Misra, K. Lindblad-Toh, C. Wade, T. Mikkelsen, E. Karlsson, D. Jaffe, M. Kamal, M. Clamp, J. Chang, E. Kulbokas, M. Zody, G. Wong, B. Liu, J. Wang, Y. Zhang, X. Yang, Z. Zhang, Q. Meng, J. Zhou, D. Li, J. Zhang, S. Eck, A. Benet-Pages, K. Flisikowski, T. Meitinger, R. Fries, T. Strom, J. Kijas, D. Townley, B. Dalrymple, M. Heaton, J. Maddox, A. McGrath, P. Wilson, R. Ingersoll, R. McCulloch, S. McWilliam, A. Ramos, R. Crooijmans, N. Affara, A. Amaral, A. L. Archibald, J. Beever, C. Bendixen, C. Churcher, et al. 2011b. Evaluation of approaches for identifying population informative markers from high density SNP chips. BMC Genet. 12:45. doi:10.1186/1471-2156-12-45.

Wright, S. 1951. The Genetical Structure of Populations. Ann. Eugen. 15:322–354. doi:10.1017/CBO9781107415324.004.

Zheng, X., and B. S. Weir. 2016a. Eigenanalysis of SNP data with an identity by descent interpretation. Theor. Popul. Biol. 107:65–76. doi:10.1016/j.tpb.2015.09.004. Available from: http://dx.doi.org/10.1016/j.tpb.2015.09.004

Zheng, X., and B. S. Weir. 2016b. Eigenanalysis of SNP data with an identity by descent interpretation. Theor. Popul. Biol. 107:65–76. doi:10.1016/j.tpb.2015.09.004. Available from: http://dx.doi.org/10.1016/j.tpb.2015.09.004

Zhivotovsky, L. A. 2015. Relationships Between Wright's Fst and Fis Statistics in a Context of Wahlund Effect. J. Hered. 106:306–309. doi:10.1093/jhered/esv019.

BIOGRAPHICAL SKETCH

Mesfin was born and raised in Addis Ababa, the capital city of Ethiopia. He studied Veterinary Medicine at Addis Ababa University, and later worked at the same university for a short period as a lecturer. Mesfin then joined Dr. Raluca Mateescu's lab at the Department of Animal Sciences, University of Florida where he worked on the genetic background of various traits in cattle, sheep and goat. He is passionate about all things science in general, and genetics, animal health and environmental adaptation in particular.